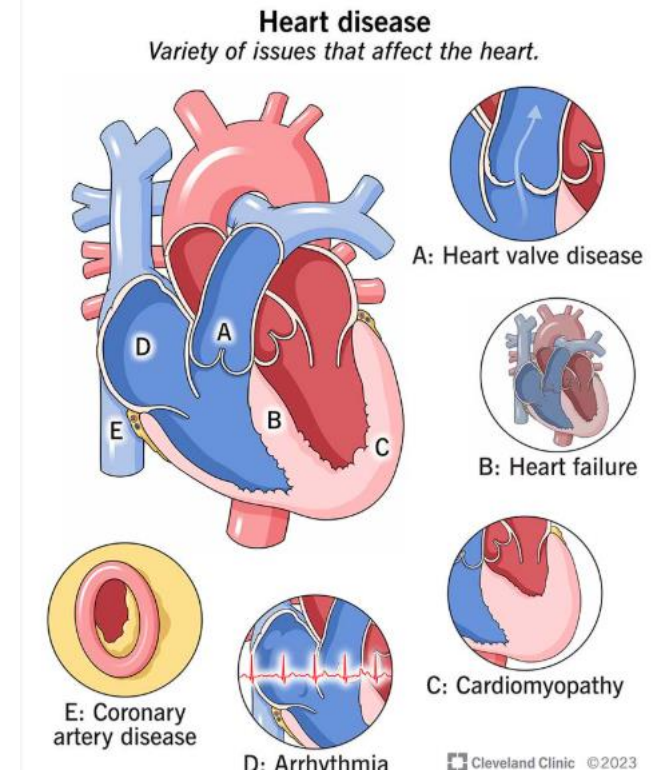


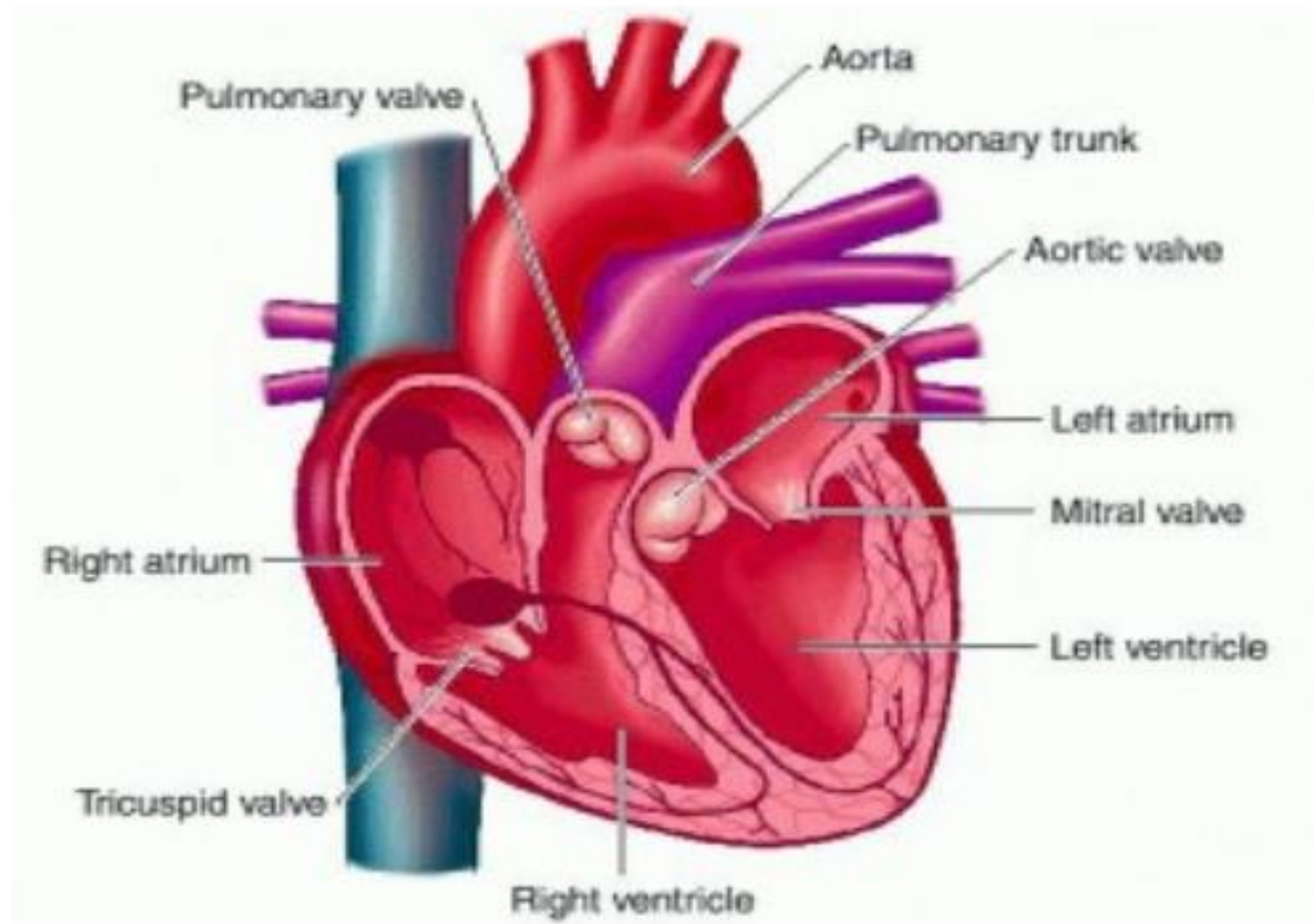
Pathology of common cardiac diseases

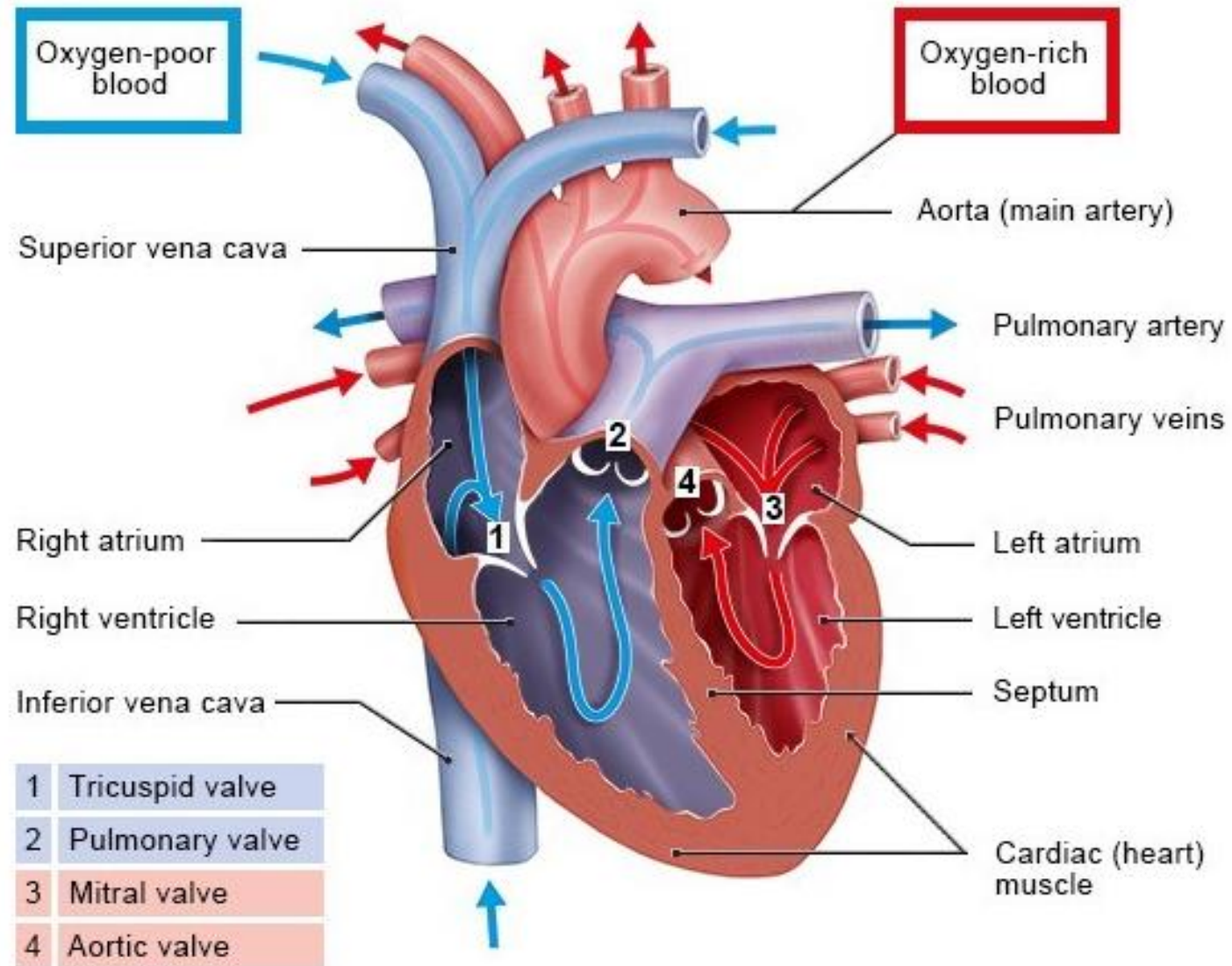
BY MELESE A.

OUTLINE

- Review of anatomy of heart
- Heart failure
- Congenital heart diseases
- Ischemic heart disease
- Valvular heart diseases
- Cardiomyopathies
- Tumors of the heart and pericardium





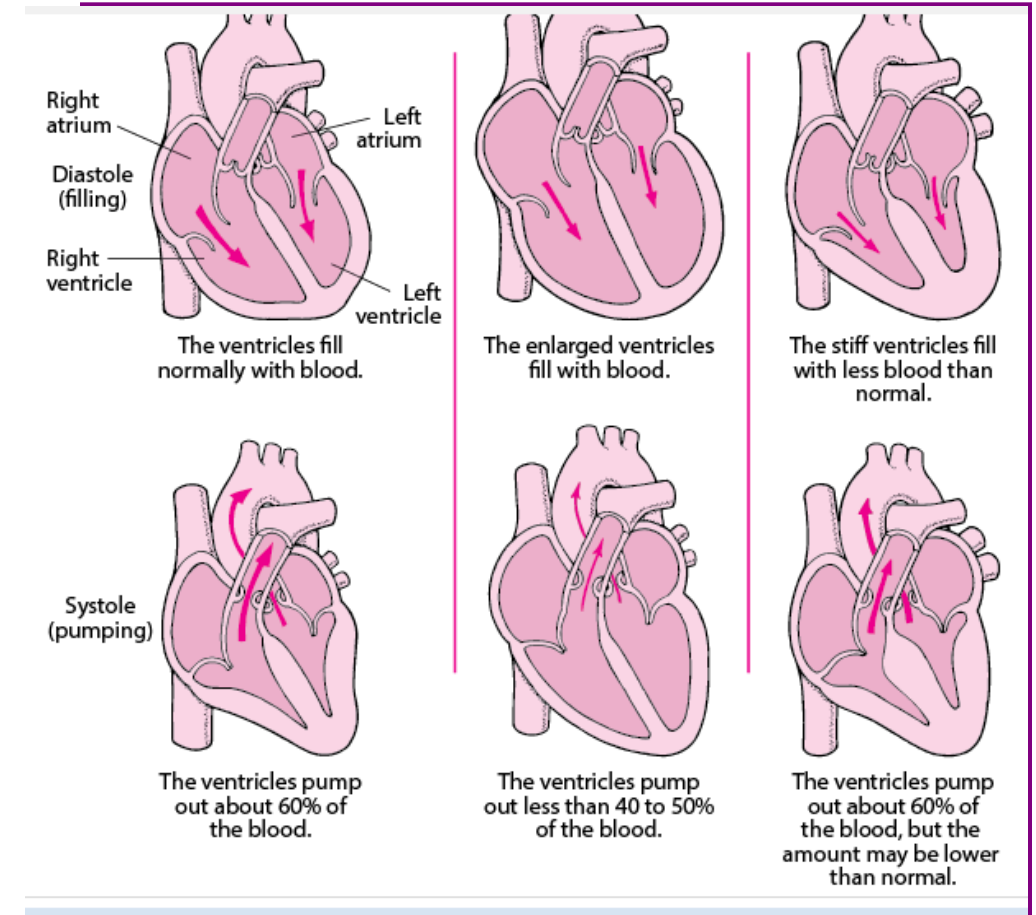


HEART FAILURE

- A clinical syndrome resulting from structural or functional cardiac disorders
- Impaired ability of the heart to **fill with** or **eject** blood adequately to meet the metabolic demands of tissues.
- The **final common pathway** of many cardiovascular diseases
- Is also called **congestive heart failure(CHF)**

Classification

- Heart failure can be a consequence of
 - **Systolic dysfunction:** Deterioration of myocardial contractile function
 - **Diastolic failure:** Relaxation is abnormal



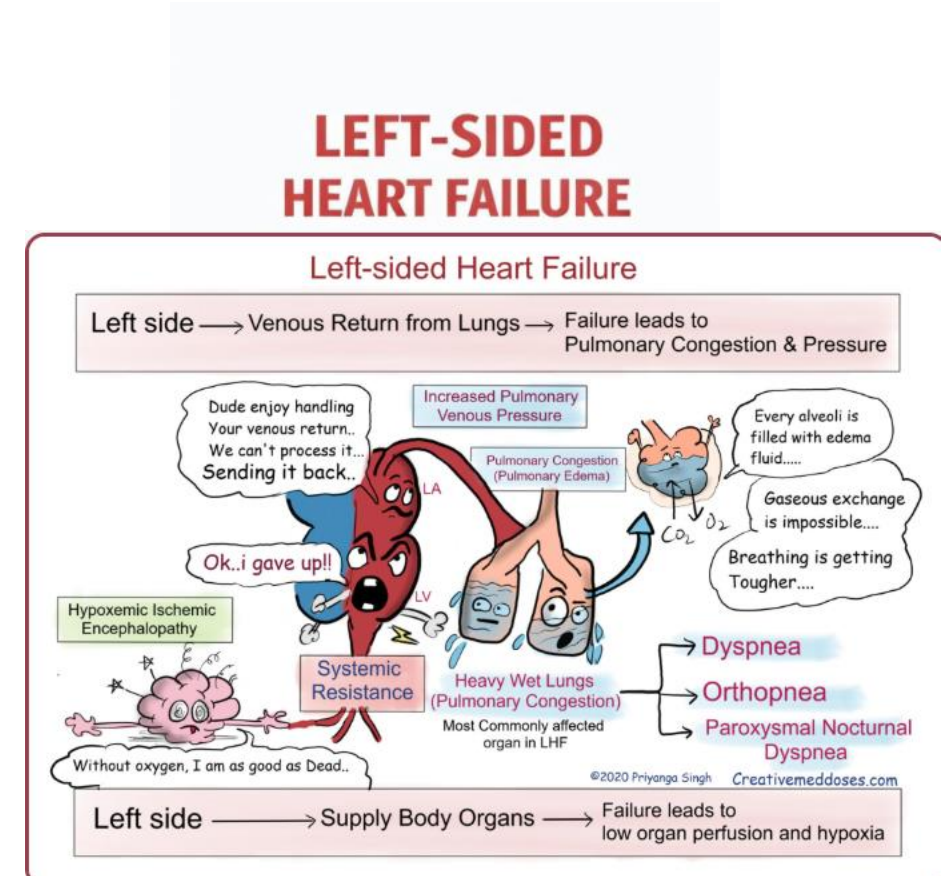
- **High-output failure**

- heart failure that results from greatly increased tissue demands for blood.
 - Severe anemia
 - Hyperthyroidism
 - Arteriovenous shunts
 - Beriberi

Left-Sided Heart Failure

- The most common causes are:
 - ❖ Ischemic Heart Disease(IHD)
 - ❖ Systemic hypertension
 - ❖ Mitral or aortic valve disease(Valvular heart diseases)
 - ❖ Primary diseases of the myocardium(Cardiomyopathies)
- Effects of left-sided CHF primarily result from
 - Progressive **damming** of blood within the **pulmonary** circulation
 - Consequences of **diminished peripheral blood pressure and flow**
- Can be divided into systolic and diastolic failure

- Findings in the heart depend on the underlying disease process.
 - ❖ Left ventricle is usually **hypertrophied** and often **dilated**
 - ❖ Secondary enlargement of the left atrium
 - ❖ Cause rising pressure in the pulmonary veins which is ultimately transmitted retrogradely to the capillaries
 - ❖ Resulting in **pulmonary congestion and edema**
 - ❖ The lungs are heavy and boggy, and histologically there are perivascular and interstitial transudate and intra-alveolar edema



• Pink Frothy Sputum

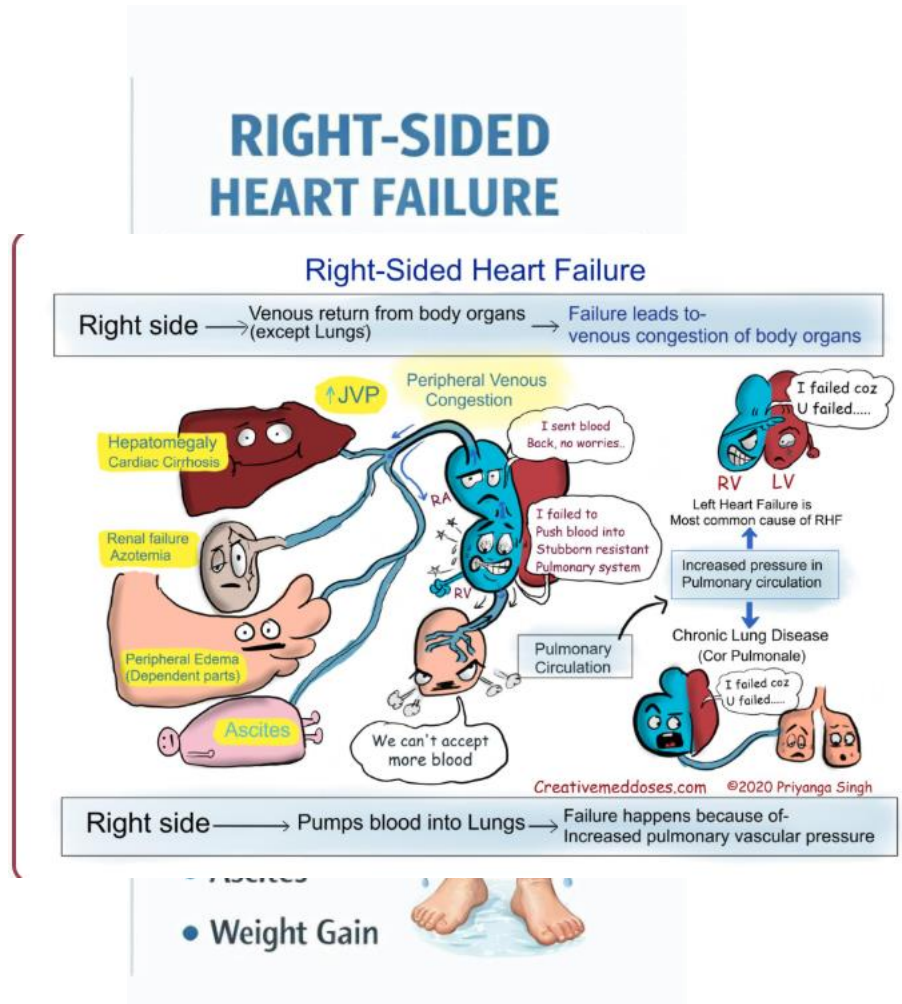
Right-Sided Heart Failure

Causes:

- Diseases of the lung parenchyma and/or pulmonary vasculature (cor pulmonale)
- Pulmonic or tricuspid valve disease
- Congenital heart diseases
- **Left-sided heart failure** - the most common cause

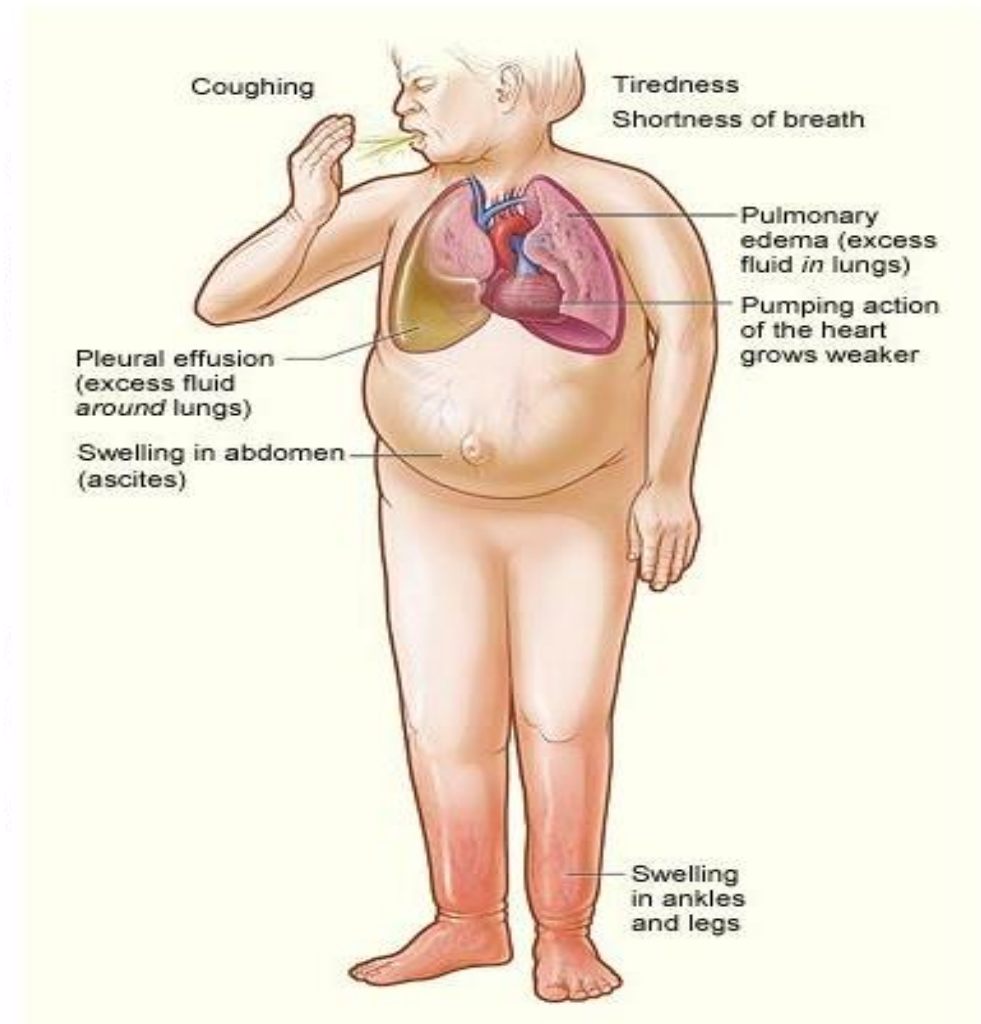
- The major morphologic and clinical effects of pure right-sided heart failure shows

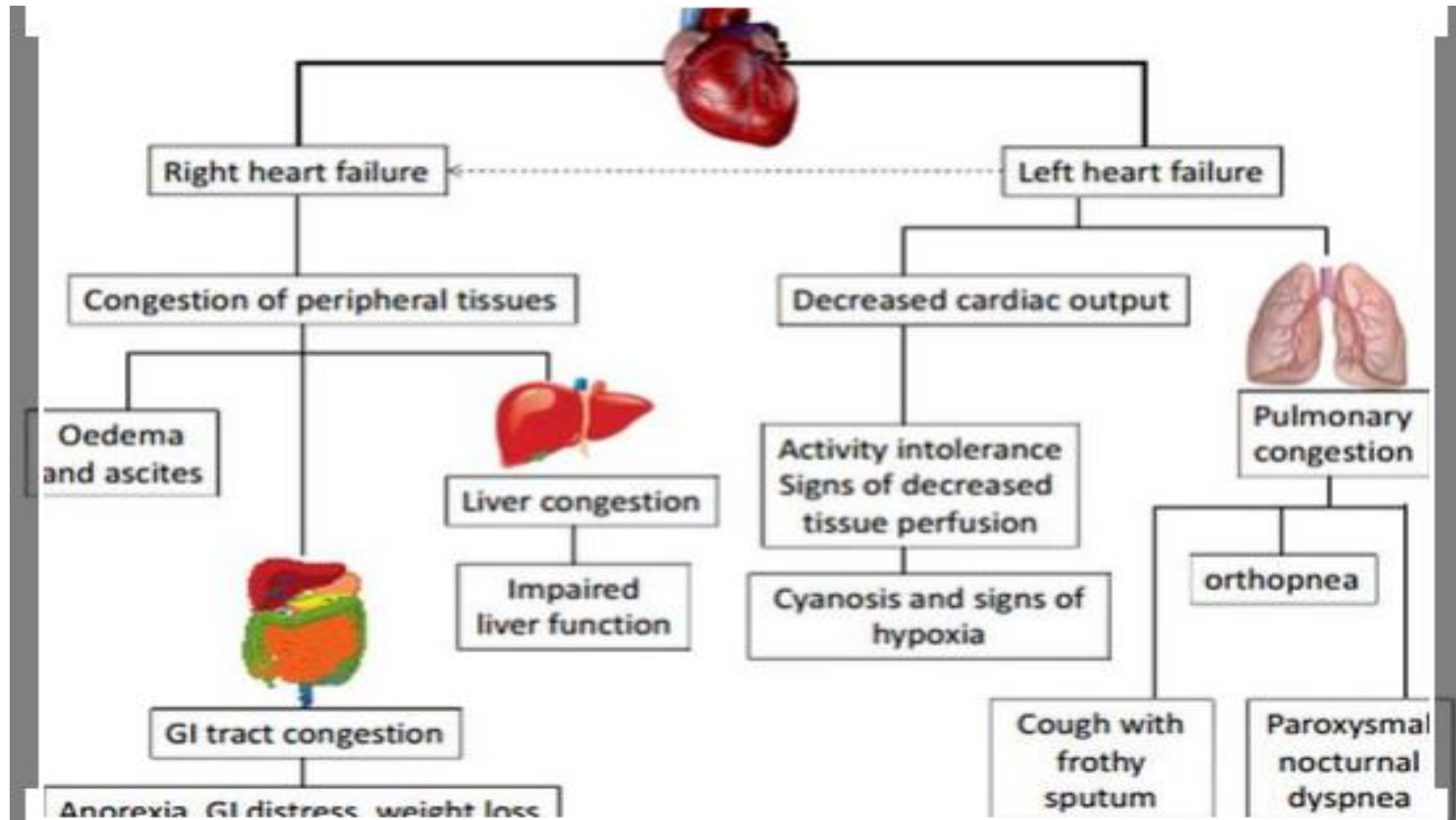
- **Minimal pulmonary congestion**
- Engorgement of the systemic and portal venous systems is typically pronounced.
 - Congestive hepatomegaly
 - Congestive splenomegaly
 - Accumulations of transudate in the peritoneal cavity can cause **ascites**



Heart Failure Signs and Symptoms

- Fluid may accumulate in the **pleural space** and **pericardial space** (effusions)
- **Subcutaneous edema:**
Peripheral edema of dependent portions of the body, especially ankle (pedal) and pretibial edema
- In bedridden patients, the edema may be primarily **presacral**





- The cardiovascular system can adapt to reduced myocardial contractility or increased hemodynamic burden by a different pathways
 1. Activation of neurohumoral systems:
 - Release of the neurotransmitter **norepinephrine** by the sympathetic nervous system
 - Activation of the **renin-angiotensin-aldosterone system**
 - Release of atrial natriuretic peptide (ANP)
 2. The Frank-Starling mechanism
 3. Myocardial adaptations

- If the dilated ventricle is able to maintain cardiac output at a level that meets the needs of the body, the patient is said to be in **compensated heart failure**
- Initially, these adaptive mechanisms may be adequate to maintain cardiac output.
- With time, the failing myocardium is no longer able to propel sufficient blood to meet the needs of the body, even at rest.
- The failing heart is accompanied by increased congestion of the venous circulation (*backward failure*)
 - the failing ventricle is unable to eject the venous blood delivered to it.

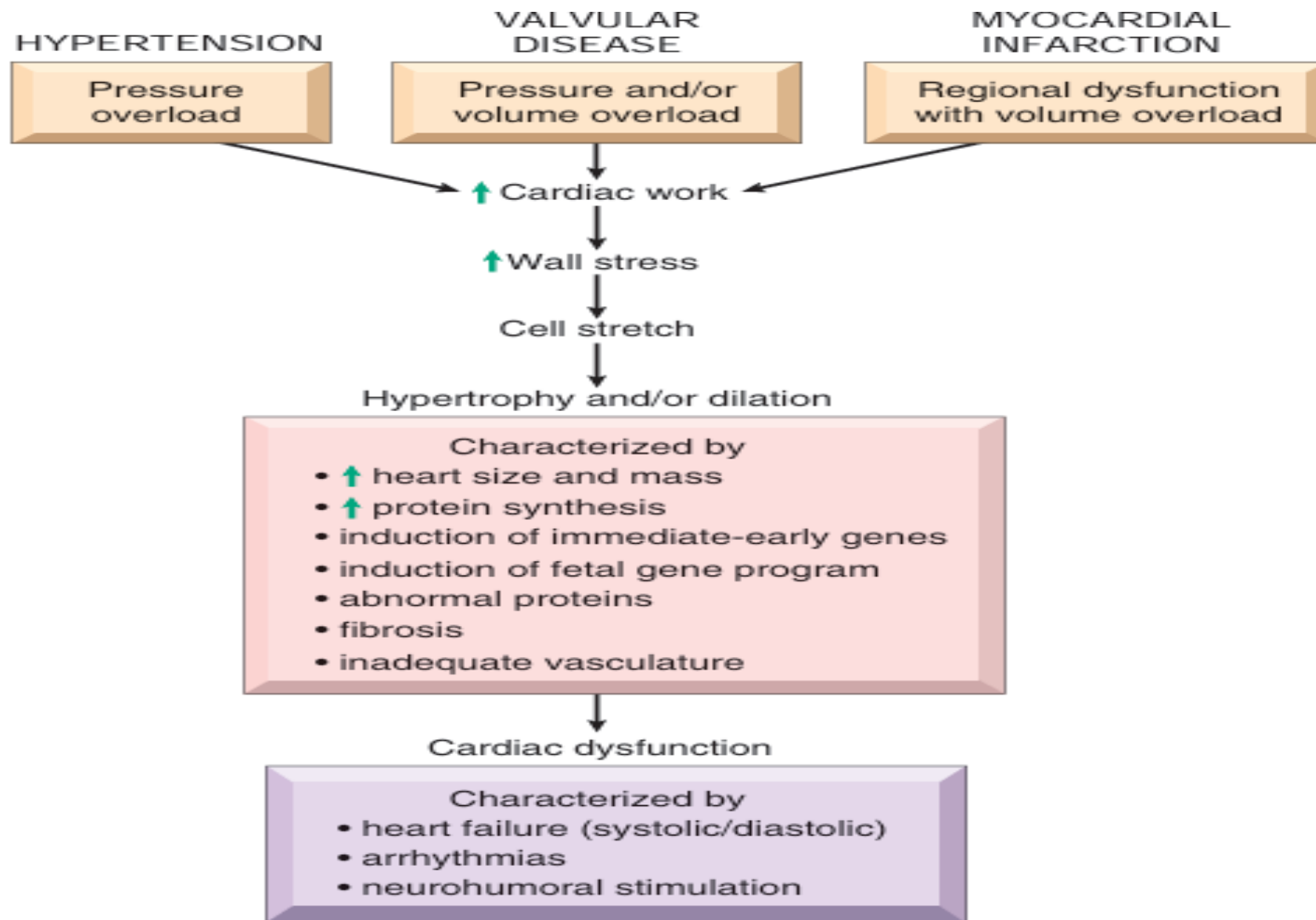


Figure 12.2 Schematic representation of the causes and consequences of cardiac hypertrophy.

NYHA Classification of CHF

CLASS-1	Patients with cardiac disease but without limitation of any physical activities.
CLASS-2	-Slight limitation of physical activities -they are comfortable at rest -Ordinary physical activities results in cvs symptoms
CLASS-3	-Marked limitation of physical activities -still they are comfortable at rest -less than ordinary activities result in cvs symptoms
CLASS -4	-Patients are unable to carry out any physical activities -symptoms occur at rest.

STAGING OF HF

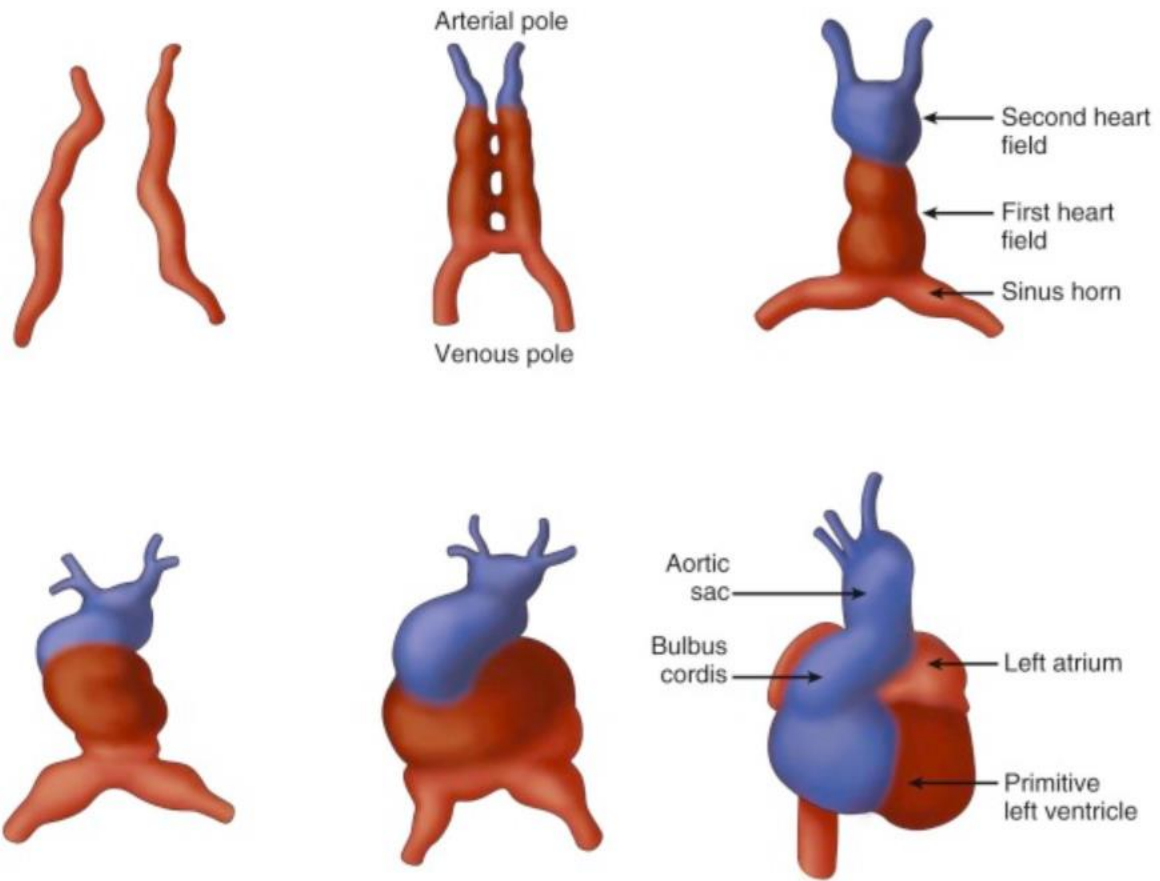
STAGE-A	The patient is predisposed to the risk factors of HF like -hypertension -coronary artery disease - DM The patient doesn't develop any symptoms of HF
STAGE-B	-Structural heart disease without symptoms of HF -Patients with previous –MI -LV systolic dysfunction -asymptomatic valvular disease
STAGE-C	Structural heart disease with symptoms of HF Dyspnea Fatigue
STAGE-D	Stage which needs specialized intervention Symptoms at rest

Causes of Death in Heart Failure

- Progressive pump failure
- Ventricular arrhythmias
- Sudden cardiac death
- Thromboembolism
- Multiorgan dysfunction

Congenital Heart Diseases (CHD)

- A spectrum of **structural and functional defects** in the heart and great vessels that are **present at birth**.
- They arise from **aberrant embryonic development** during the first 3 - 8 weeks of gestation,
- Occur in 0.5 – 0.8% of live births
- Etiology: unknown in most cases
 - The cause of most CHDs is **multifactorial**
 - **Genetic Factors:** Chromosomal anomalies (Trisomy 21, 22q11.2 deletion syndrome), single-gene mutations (e.g., *NKX2-5*).
 - **Environmental Teratogens:** Maternal rubella infection, uncontrolled diabetes mellitus, alcohol use, and certain medications (e.g., lithium, isotretinoin).
- Can be classified based on their hemodynamic consequences
 - **shunting of blood (left-to-right or right-to-left)**
 - **obstruction to flow**



Classification of CHD

Cyanotic Heart Disease

- Decreased pulmonary flow:
 - Tetralogy of Fallot
 - Tricuspid atresia
 - Other univentricular heart with pulmonary stenosis.
- Increased pulmonary flow:
 - Transposition of great arteries
 - Total anomalous pulmonary venous return.

Acyanotic Heart Disease

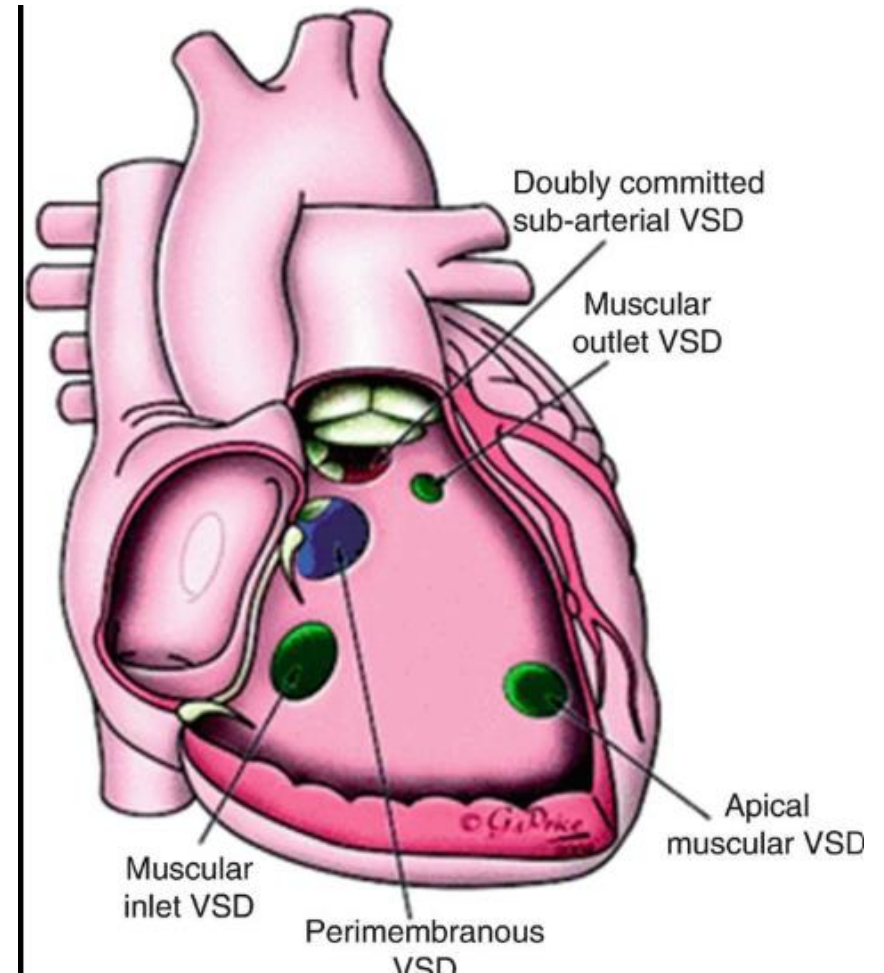
- Left – Right shunt lesions:
 - Ventricular septal defect
 - Atrial Septal Defect
 - Atrio-ventricular Septal Defect
 - Patent Ductus Arteriosus
- Obstructive lesions:
 - Aortic stenosis
 - Pulmonary valve stenosis
 - Coarctation of Aorta

Ventricular Septal Defect (VSD)

- A hole in the interventricular septum.
- The most common CHD
- Two types
 - Membranous (most common)
 - Muscular

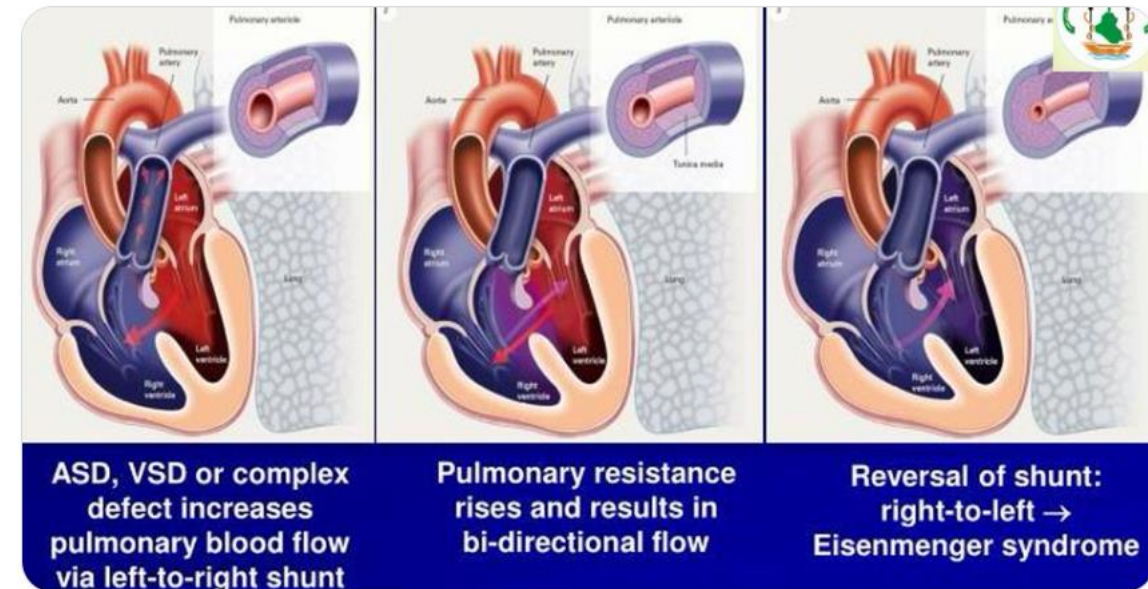
Pathophysiology

- Shunting of blood from L to R ventricle
- L to R shunt depend on:
 - Size of VSD
 - Pulmonary vascular resistance (PVR)



VSD

- In small VSD right V pressure is normal
→ high pressure gradient from L to R
- In large VSD there is large L to R shunt
 - Symptoms appear
 - Increased PVR
- If pulmonary and systemic pressures equalize → clinical symptoms improve
- Later when right V pressure exceeds that of left → patient becomes **cyanotic**
 - Eisenmenger syndrome



Clinical Manifestation

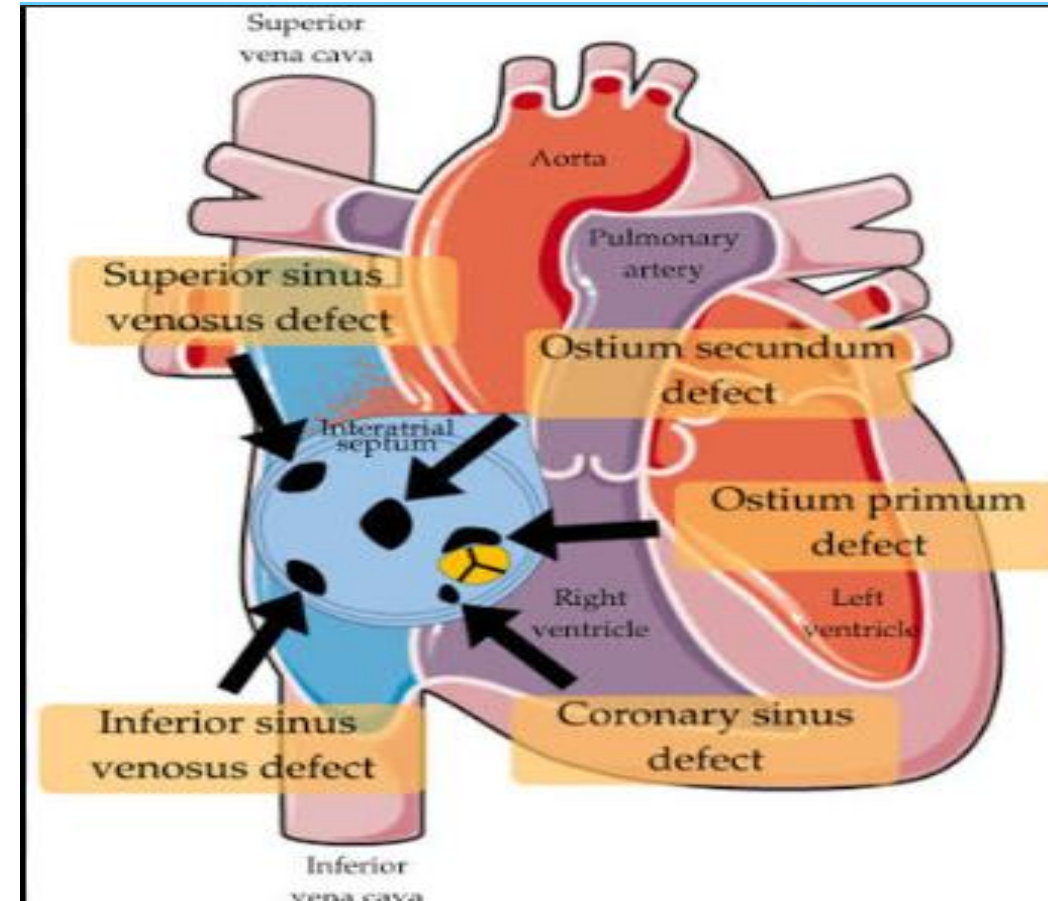
- Varies according to:
 - Size of the defect
 - Pulmonary blood flow & pressure
- **In small VSD (most common):**
 - Patients are asymptomatic
 - Characteristically there is a loud, harsh or blowing **holosystolic** murmur (over LLSB)
- **In large VSD:**
 - Patients are symptomatic with dyspnea, feeding difficulties, poor growth, facial diaphoresis, recurrent respiratory infections and congestive heart failure
 - N.B – murmur of large VSD is **less harsh** than small VSD

Atrial Septal Defect

- Defect occur in any portion of the atrium
 - **Ostium Secundum (most common):** Defect in the fossa ovalis region.
 - **Ostium Primum:** Defect low in the septum.
 - **Sinus Venosus:** Defect high in the septum

Pathophysiology

- Left to right shunt
 - Transatrial in OS & SV
 - Transatrial & transventricular in OP
- ASDs are generally well tolerated and usually do not become symptomatic before 30 years of age



Atrial Septal Defect

Morphology

- Right atrial enlargement
- Right ventricular dilation
- Pulmonary artery enlargement

Complications:

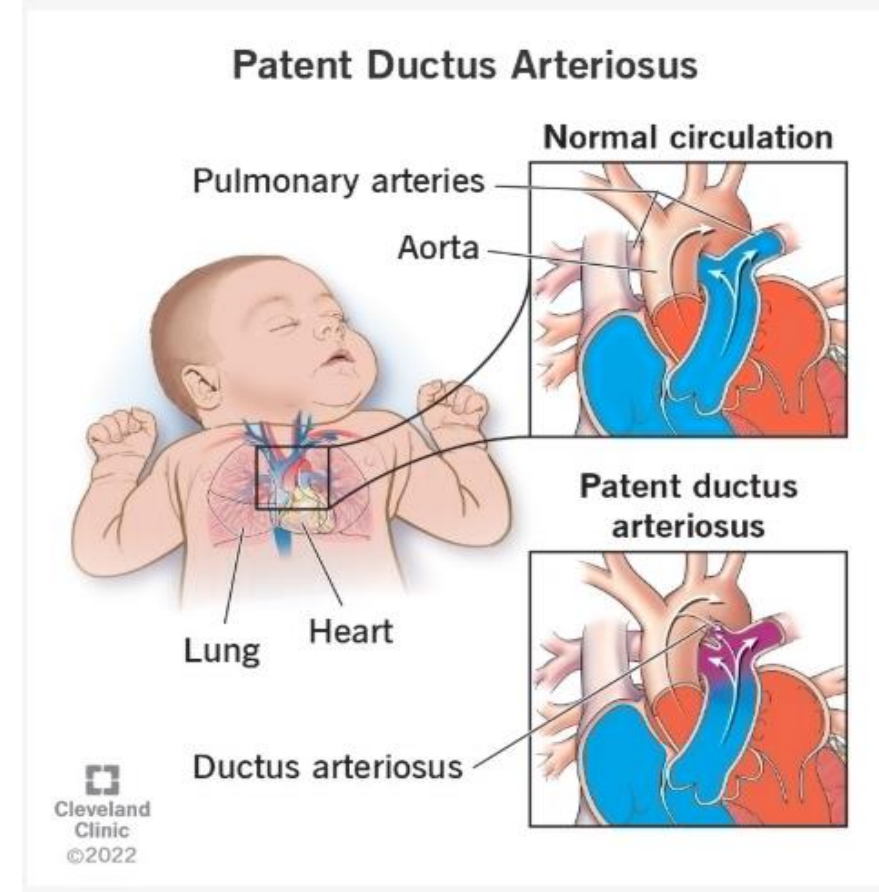
- Arrhythmias
- Pulmonary hypertension

Patent Ductus Arteriosus (PDA)

- Failure of the ductus arteriosus to close after birth.

Pathophysiology:

- After birth, aortic pressure exceeds pulmonary artery pressure.
- Blood shunts from the aorta to the pulmonary artery (*a left-to-right shunt*),
 - increasing pulmonary blood flow and returning to the left heart
 - causing **volume overload** of the left atrium and ventricle.

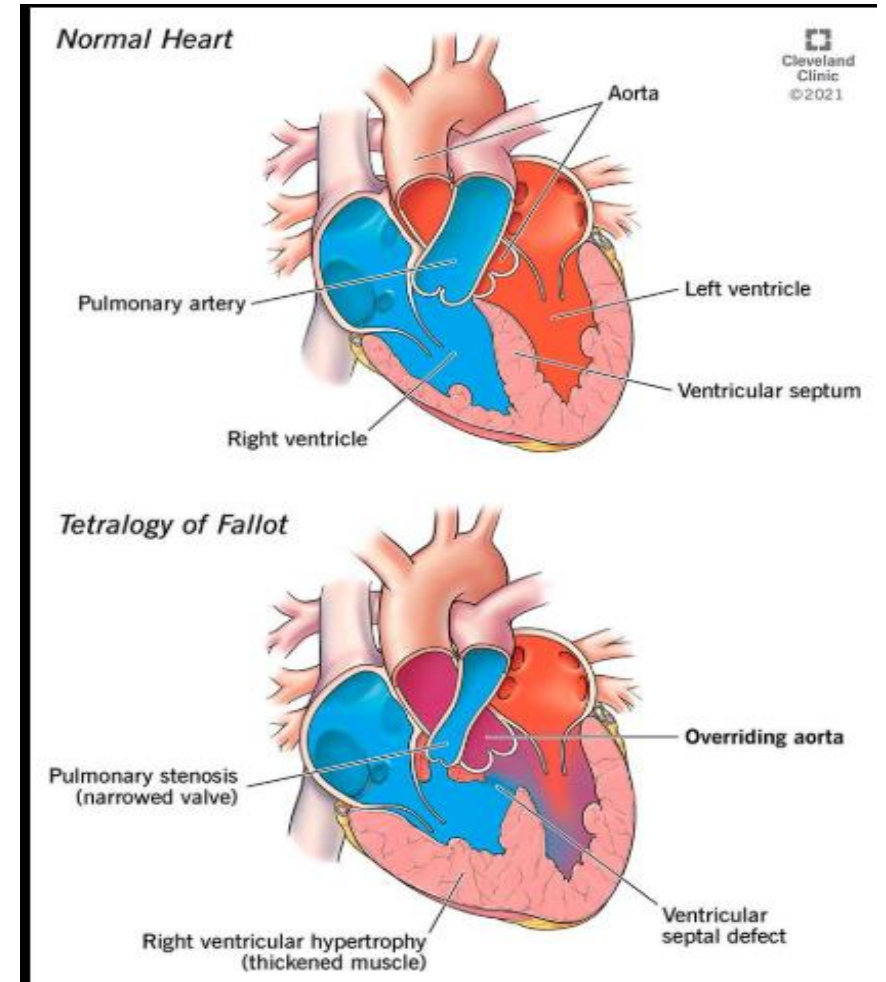


Tetralogy of Fallot (TOF)

- Cyanotic CHD
- Consists of:
 - Pulmonary stenosis
 - VSD
 - Overriding aorta
 - Right ventricular hypertrophy

Clinical manifestation

- In infants with mild degree of right V outflow obstruction (RVOO) manifest with:
 - Initially with congestive heart failure (CHF)
 - Often cyanosis not present at birth



TOF

- In severe degree of RVOO:
=> Cyanosis noted immediately
- Pulmonary blood flow in the first few days depend on PDA
- When ductus closes → severe cyanosis and circulatory collapse
- Dyspnea on exertion (assume SQUATTING position)
- Characteristic paroxysmal hypercyanotic attacks (hypoxic, 'blue' or 'tet' spells)
- Spells are common in the first 2yrs of life
- Infant becomes hyperpneic and restless, cyanosis increases; and has gasping respiration and syncope
- Spells occur in the morning on awakening or follow vigorous crying

Tetralogy of Fallot

it is the most common form of cyanotic congenital heart disease.

- Ventricular septal defect
- Pulmonary Stenosis
- Right ventricular hypertrophy
- Overriding aorta

Clinical features :

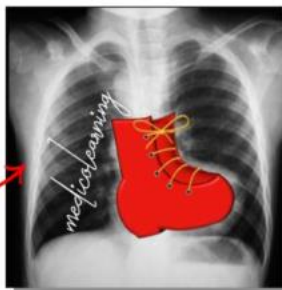
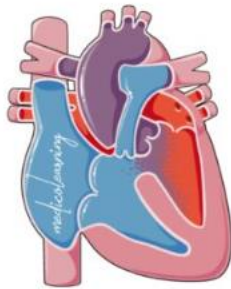
- Cyanotic Spells
- Clubbing of fingers
- Poor development
- Squatting during episodes of cyanosis
- Metabolic acidosis
- Fatigue and dyspnoea

Investigations:

- ECG: Right ventricular hypertrophy
- Chest X ray: shows BOOT shaped heart
- ECHO confirms the diagnosis

Management:

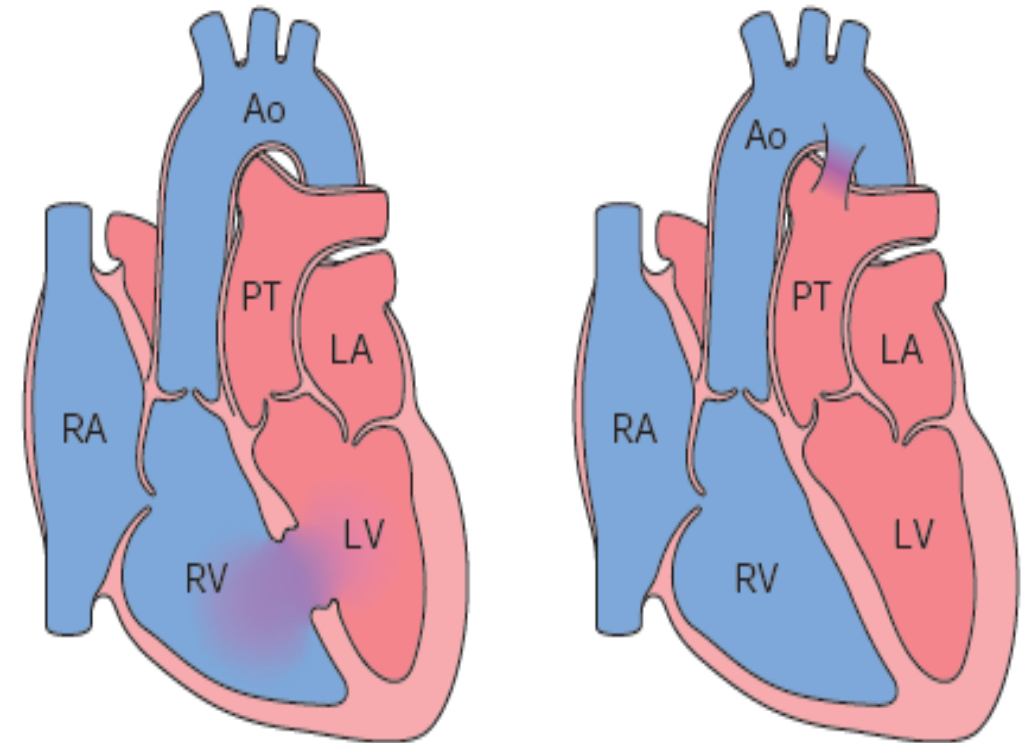
- Complete correction in infants and children.
- Blalock-Taussig shunt
- Cyanotic spells are treated by:
 - knee-to-chest" position
 - Administering oxygen.
 - I.V fluids to increase venous return
 - Morphine 0.1 mg/kg i.v
 - Antibiotic prophylaxis for endocarditis.



Flash Card

Transposition of Great Arteries (TGA)

- Cyanotic CHD
- Aorta arises from RV and pulmonary artery from LV
- Systemic and pulmonary circulation consist of two parallel circuits
- Survival is provided by patent foramen ovale and PDA
- Half of patients with TGA have VSD



With VSD

Without VSD

B Complete transposition

- Common in infants of **diabetic** mothers
- Prior to modern surgical treatment mortality was greater than 90%
- Clinical manifestation
 - Cyanosis with in the first hours or days of life
 - A medical emergency
 - Early diagnosis and appropriate intervention can avert severe hypoxemia and acidosis which leads to death

Persistent Truncus Arteriosus

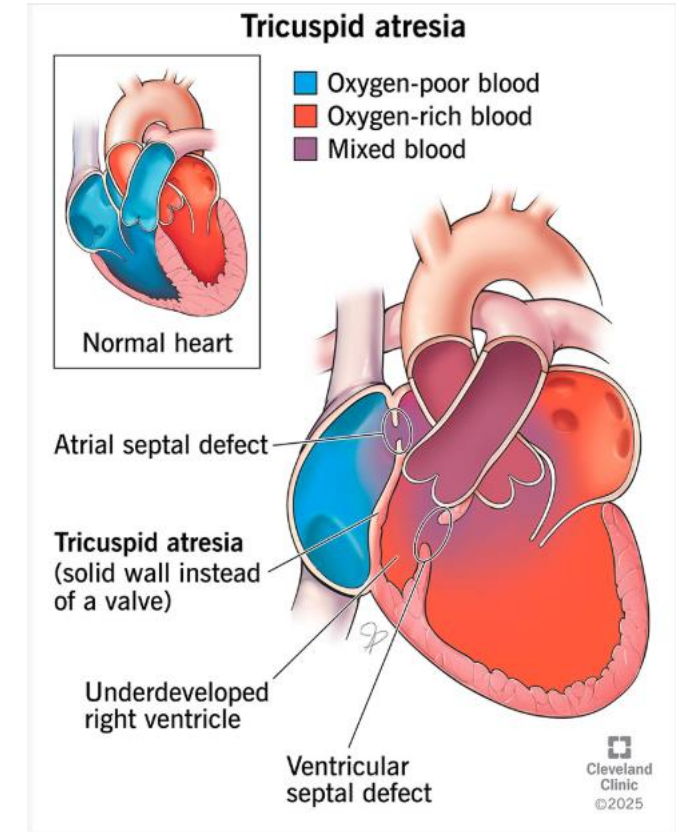
- Failure of truncal septation.
- Single arterial trunk supplies:
 - Systemic circulation
 - Pulmonary circulation
 - Coronary circulation

Total Anomalous Pulmonary Venous Return (TAPVR)

- Pulmonary veins connect to systemic venous circulation rather than left atrium.
- Results:
 - Mixing of oxygenated and deoxygenated blood
 - Cyanosis

Tricuspid Atresia

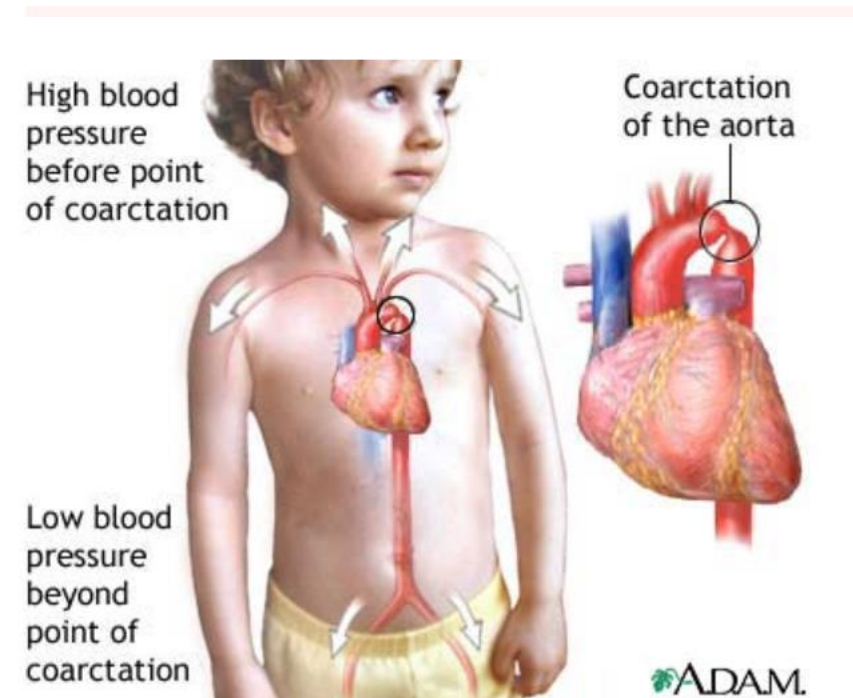
- Complete agenesis of the tricuspid valve.
- No communication exists between the right atrium and right ventricle.
- Blood from the right atrium must shunt **right-to-left** through an ASD to the left atrium.
- The left ventricle pumps blood to the body and, via a VSD (if present), to a small, underdeveloped right ventricle and then to the lungs.
- Cyanosis is always present.



Obstructive Lesions

Coarctation of the Aorta

- A localized narrowing of the aortic lumen
 - distal to the origin of the left subclavian artery near the ligamentum arteriosum.
- Creates two physiologic patterns
 - **Preductal (Infantile type):** Severe narrowing proximal to the PDA.
 - Survival depends on the PDA remaining open to supply blood to the lower body.
 - Presents with heart failure and shock when the PDA closes.
 - **Postductal (Adult type):** Collateral circulation develops (e.g., via intercostal arteries),
 - hypertension is present in the upper extremities with weak pulses and hypotension in the lower extremities.
 - Leads to left ventricular hypertrophy.



Pulmonary Stenosis / Aortic Stenosis

- Narrowing at the valvular, subvalvular, or supravalvular level.
- A **bicuspid aortic valve** is a frequent cause of aortic stenosis in adults,
 - abnormal valve is prone to fibrosis and calcification.
- The respective ventricle must generate higher pressure to eject blood, leading to pressure overload and concentric hypertrophy.
- Severe stenosis can lead to heart failure.

Consequences and Complications of CHD

- **Heart Failure:** Due to chronic volume or pressure overload.
- **Infective Endocarditis:** bacterial colonization of defective valves.
- **Eisenmenger's Syndrome:** The irreversible shunt reversal
- **Paradoxical Embolism:** An embolus from the venous circulation can cross a shunt (ASD, VSD) into the systemic arterial circulation
- **Hypoxemia and Cyanosis:** Leading to clubbing of digits and polycythemia

ISCHEMIC HEART DISEASE

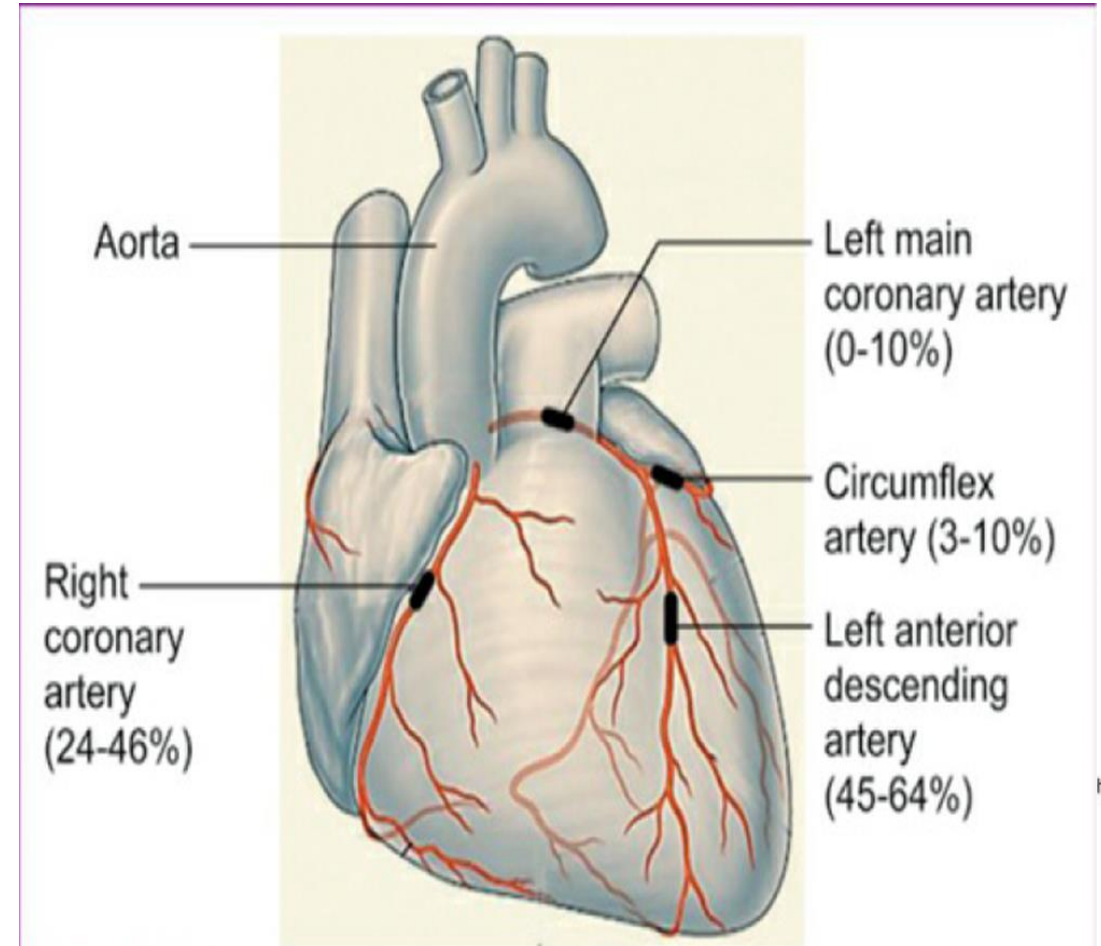
- Occurs when there is insufficient blood supply to the myocardium to meet functional demand
- A balance between myocardial blood **supply** and myocardial **demand** for oxygen and other nutrients is altered
- Can be classified as **atheroscleorotic and non atherosclerotic**
- Ischemic heart disease comprises:
 - **Coronary atherosclerosis**
 - **Non atherosclerotic**
 - hypertensive heart disease, aortic valve disease
 - anomalies of the coronary circulation, polyarteritis
 - cardiomyopathies, congenital heart disease

TABLE 6.4 Risk factors for ischaemic heart disease

Modifiable	Non-modifiable
Cigarette smoking	Age
Hyperlipidaemias	Male sex
Hypertension	Family history
Diabetes	Personality type
High-density lipoprotein levels	
Obesity	
Exercise level	

The Coronary Arteries and Ischemic Heart Disease

- In more than 90% of cases, IHD is a consequence of **reduced coronary blood flow** secondary to **obstructive atherosclerotic vascular disease**
- Inadequate coronary perfusion relative to myocardial demand as a consequence of
 - A preexisting (“fixed”) atherosclerotic occlusion
 - New superimposed thrombosis and/or vasospasm



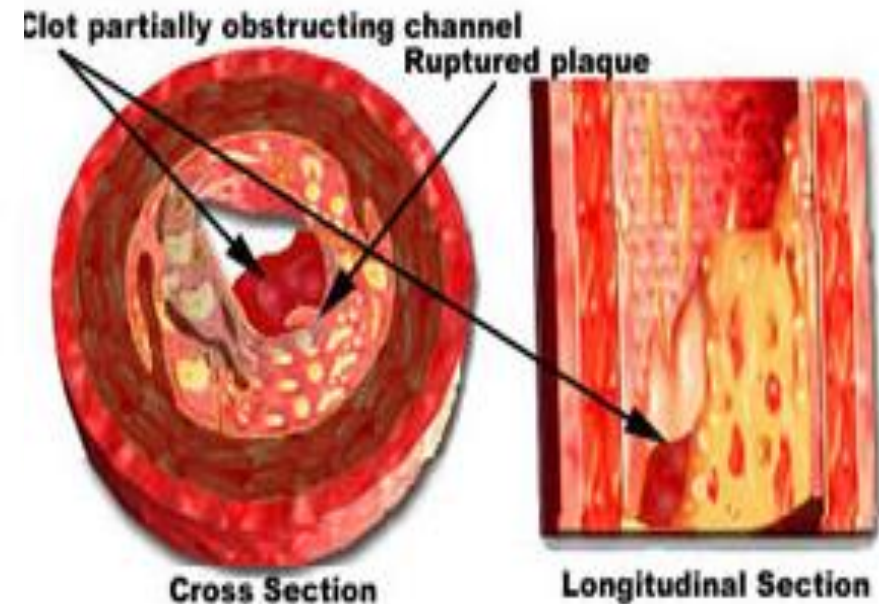
Atherosclerosis is a chronic, inflammatory, fibroproliferative disease of the arterial wall.

Pathogenesis (Response-to-Injury Hypothesis):

- Endothelial injury:
- Lipid accumulation:
- Monocyte recruitment & foam cell formation:
- Smooth muscle cell (SMC) proliferation & matrix synthesis:
- Inflammation & plaque growth:

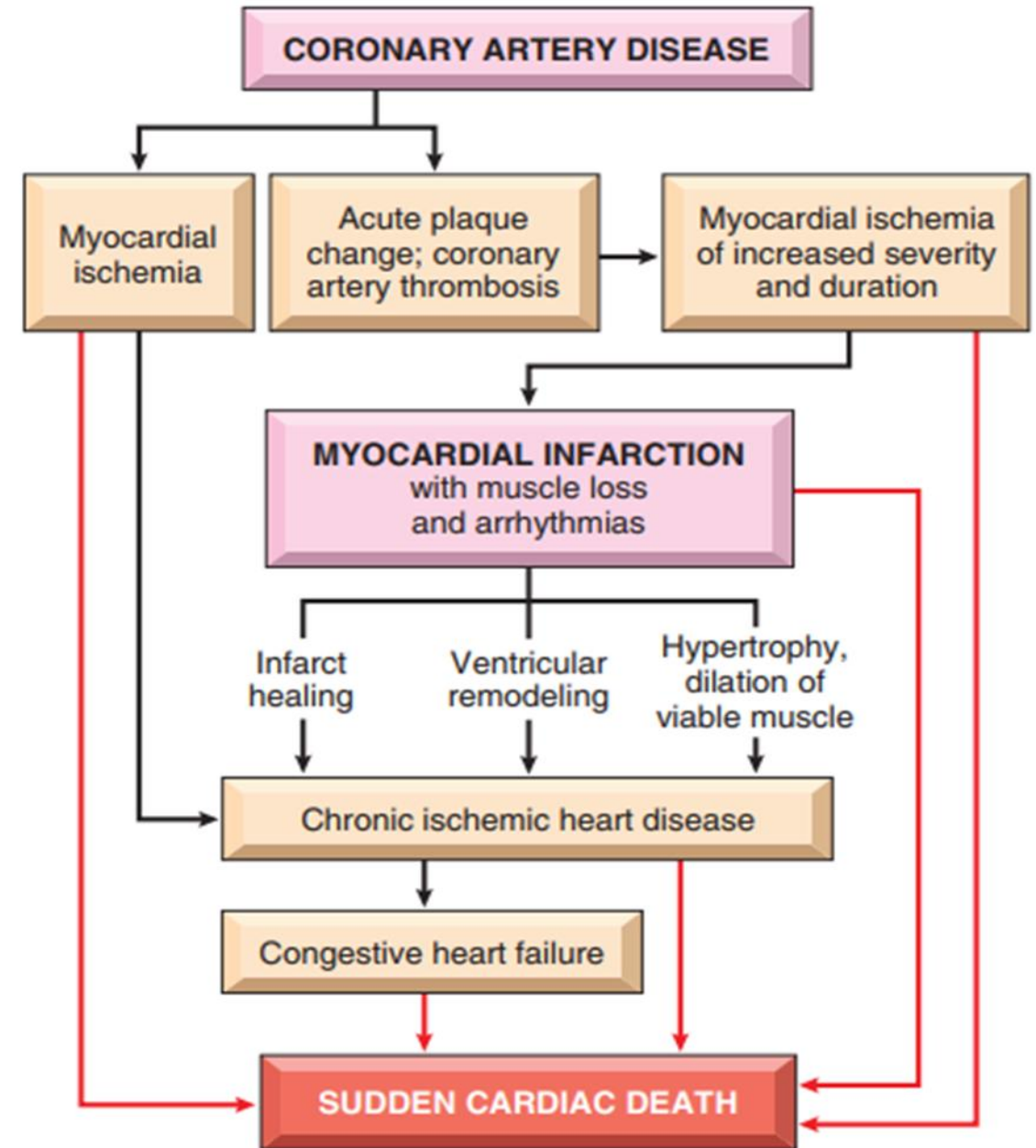
Occlusion stenosis:

- Chronic stable angina
- Plaque rupture
 - Partial occlusion: UA,NSTEMI
 - Total occlusion : STEMI



Spectrum of IHD

- Chronic stable angina
 - Fixed, stable atherosclerotic plaque causing a **critical stenosis** (typically >70% narrowing)
- Acute coronary syndrome: Due to plaque rupture or erosion
 - Unstable angina
 - The thrombus is non-occlusive (partial blockage) or transient.
 - Non ST – elevation MI
 - The thrombus causes a subtotal occlusion.
 - ST elevation MI
 - The thrombus causes complete, sustained occlusion of the coronary artery.
 - This leads to **transmural** infarction
- Chronic IHD with CHF
- Sudden cardiac death (SCD)



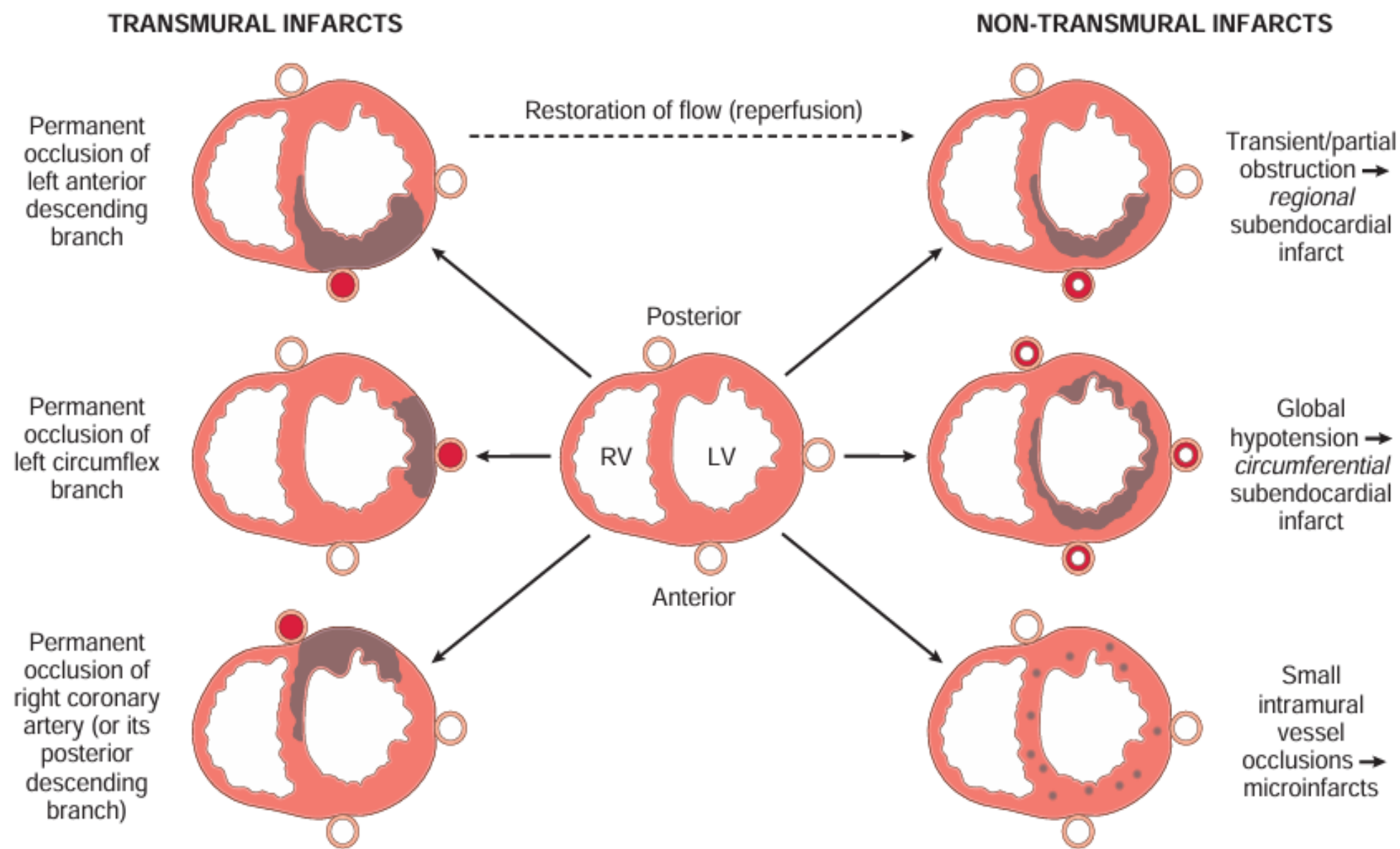
Myocardial Infarction

- Is necrosis of the heart muscle resulting from ischemia
- 45% occur before 65 years of age
- The vast majority of MIs are caused by **acute thrombosis** within coronary arteries
- The original lumen must be reduced to **20%** or less before the ischemia is sufficient to cause myocardial necrosis

Table 12.4 Approximate Time of Onset of Key Events in Ischemic Cardiac Myocytes

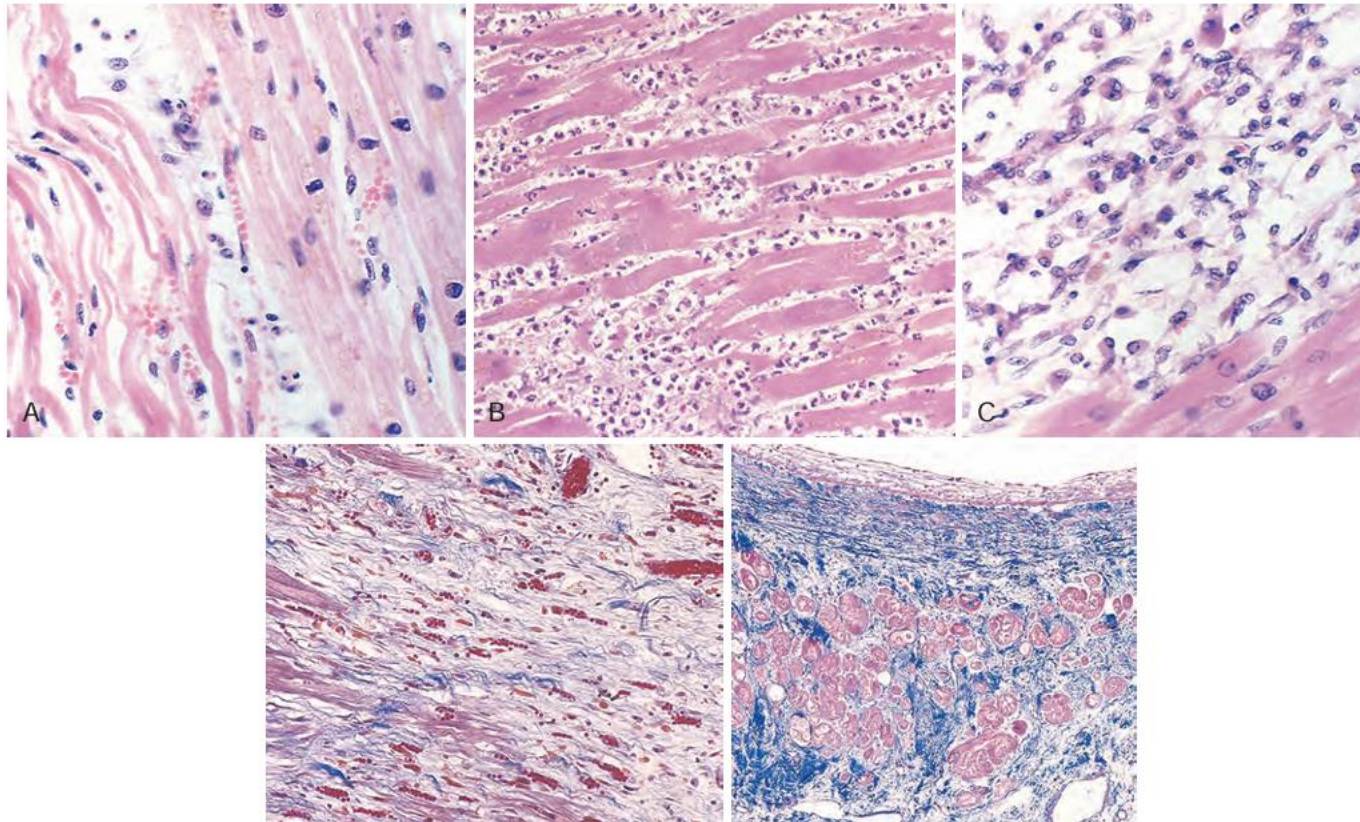
Feature	Time
Onset of ATP depletion	Seconds
Loss of contractility	<2 minutes
ATP reduced to 50% of normal	10 minutes
to 10% of normal	40 minutes
Irreversible cell injury	20–40 minutes
Microvascular injury	>1 hour

ATP, Adenosine triphosphate.



Time after onset of symptoms (approx.)	Macroscopic changes	Microscopic changes
Up to 18 hours	None	None
24 - 48 hours	Pale, oedematous muscle	Oedema, neutrophil infiltration, necrosis of myocytes
3 - 7 days	Yellow, rubbery centre with haemorrhagic border	Obvious necrosis and inflammation: early granulation tissue
1 - 3 weeks	Infarcted area paler and thinner than unaffected ventricle	Granulation tissue then progressive fibrosis
3 - 6 weeks	Silvery scar becoming tough and white	Dense fibrosis

Microscopic features of myocardial infarction and its repair



Clinical manifestations

- Prolonged retrosternal pain usually lasting more than 30 min
- Profound weakness, dyspnea
- Nausea & vomiting
- Sweating
- Asymptomatic (25%)

ECG

- ST segment elevation
- ST segment depression
- later appearance of Q wave

Markers of cardiac damage

- CK, Myoglobin, Troponins (cTnT and cTnI)

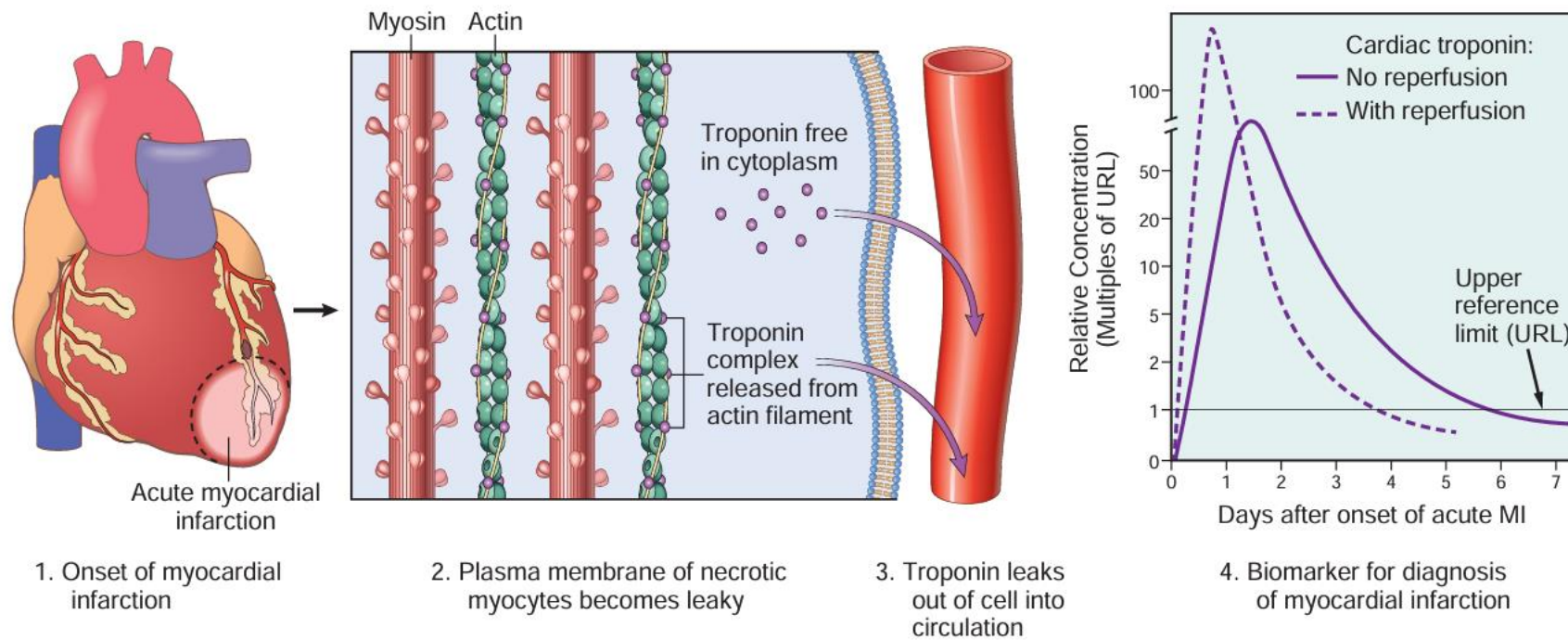
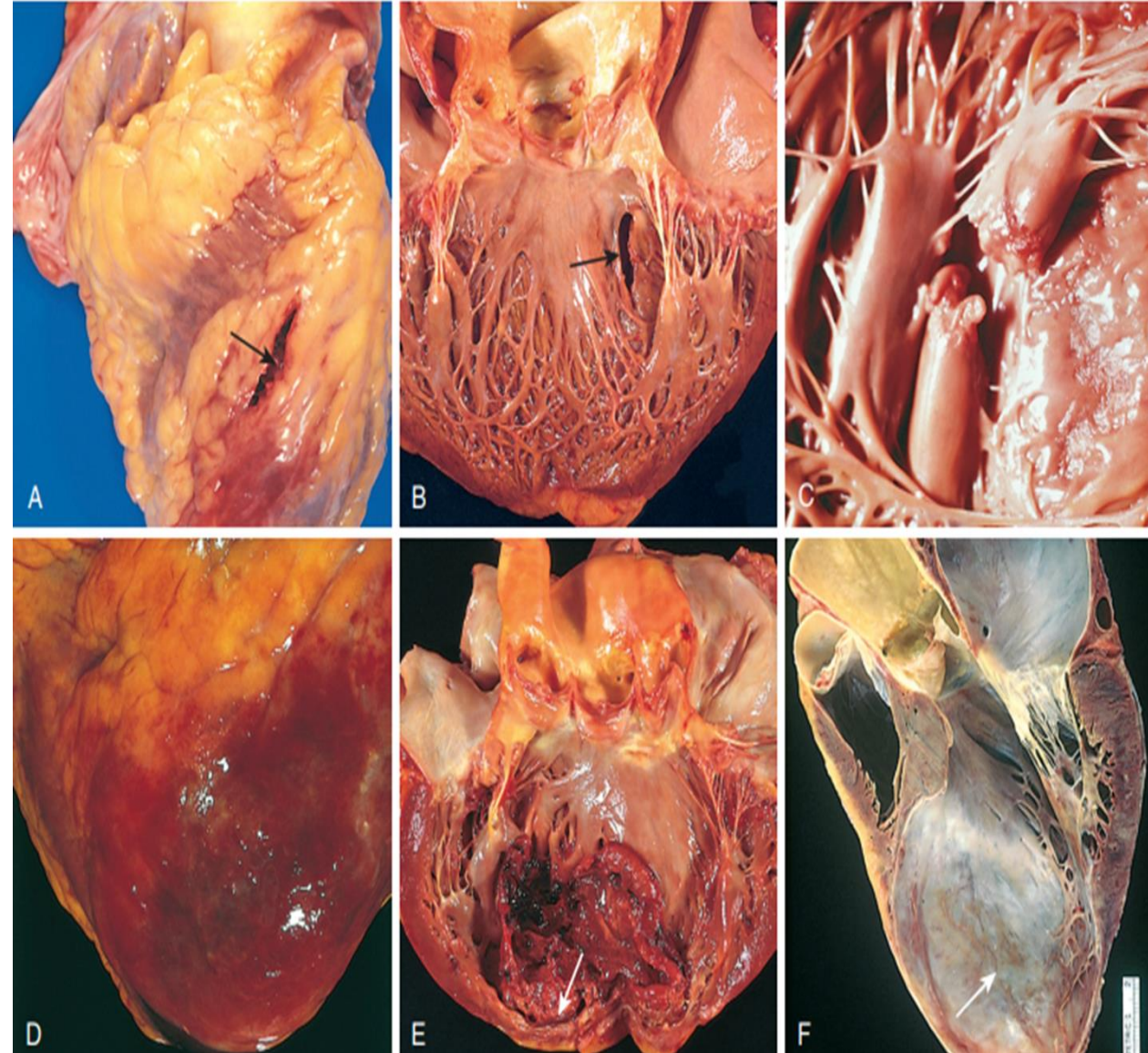


Figure 12.16 Release of myocyte proteins in myocardial infarction. Troponin I or troponin T are now most routinely used as diagnostic biomarkers of myocyte injury.

Complications of MI

- Ruptured heart – cardiac tamponade
- Mural thrombosis
- Pericarditis – Dresslers syndrome
- Myocardial fibrosis
- Cardiogenic shock



ARRHYTHMIAS

- Cardiac arrhythmias are abnormalities in the generation, conduction, or propagation of electrical impulses within the heart.
- Result in disturbances of heart rate, rhythm, or both.
- Aberrant rhythms can be initiated anywhere in the conduction system
 - Atrium (supraventricular)
 - Aentricular myocardium
- Can manifest as
 - tachycardia (fast heart rate)
 - bradycardia (slow heart rate)
 - an irregular rhythm with normal ventricular contraction
 - chaotic depolarization without functional ventricular contraction (ventricular fibrillation)
 - no electrical activity at all (asystole)

- Ischemic injury is a common cause of rhythm disorders
 - Sick sinus syndrome, Atrial fibrillation, Heart block
- Ischemia, myocyte hypertrophy, and inflammation promote increased “irritability”
- **Channelopathies** are the most important of the primary electrical abnormalities
 - Na⁺, K⁺, and Ca⁺ channels

Sudden Cardiac Death

- **Unexpected death** from cardiac causes, either
 - without symptoms or
 - within 1 to 24 hours of symptom onset
- Most often due to a **lethal arrhythmia** (e.g., asystole or ventricular fibrillation).
- CAD is the leading cause of SCD
 - It can be the **first manifestation** of IHD
- Nonatherosclerotic causes (In younger victims)
 - Cardiac conduction abnormalities,
 - Dilated or hypertrophic cardiomyopathy
 - Congenital coronary arterial abnormalities
 - Myocarditis
 - Mitral valve prolapse
 - Pulmonary hypertension

HYPERTENSIVE HEART DISEASE

- Is a consequence of the increased demands placed on the heart by **hypertension**

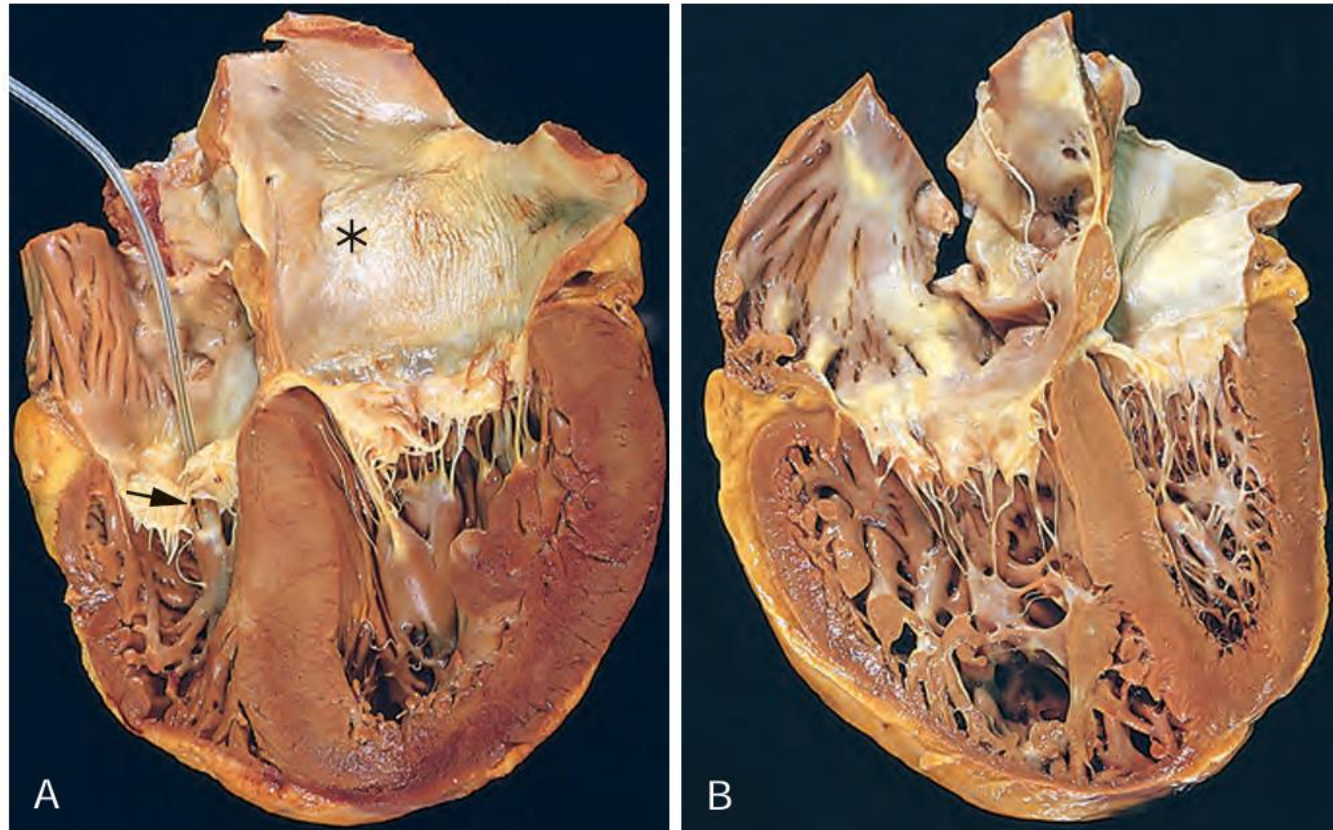
A. Systemic (Left-Sided) Hypertensive Heart Disease

- Pressure overload of **chronic hypertension**
 - Left ventricular hypertrophy (usually concentric)
 - impaired diastolic filling, often leading to left atrial enlargement

B. Pulmonary (Right-Sided) Hypertensive Heart Disease (Cor Pulmonale)

- Stems from right ventricular pressure overload
 - right ventricular hypertrophy, dilation, and potentially right-sided failure
- Causes
 - emphysema, primary pulmonary hypertension
 - massive pulmonary embolism
 - left-sided heart disease

Hypertensive heart disease



VALVULAR HEART DISEASE

- A diseased heart valve is:
 - One that fails to close(Regurgitation)
 - One that fails to open fully(Stenosis)
 - Combination
- May be congenital or acquired
- In developing countries **rheumatic valve disease** is the major cause
- In developed countries **degenerative calcification of the aortic valve** and **degenerate myxoid change in the mitral valve** predominate

Table 12.8 Etiology of Acquired Heart Valve Disease

Mitral Valve Disease	Aortic Valve Disease
Mitral Stenosis	Aortic Stenosis
Postinflammatory scarring (rheumatic heart disease)	Postinflammatory scarring (rheumatic heart disease) Senile calcific aortic stenosis Calcification of congenitally deformed valve
Mitral Regurgitation	Aortic Regurgitation
Abnormalities of leaflets and commissures Postinflammatory scarring Infective endocarditis Mitral valve prolapse “Fen-phen”—induced valvular fibrosis Abnormalities of tensor apparatus Rupture of papillary muscle Papillary muscle dysfunction (fibrosis) Rupture of chordae tendineae Abnormalities of left ventricular cavity and/or annulus Left ventricular enlargement (myocarditis, dilated cardiomyopathy) Calcification of mitral ring	Intrinsic valvular disease Postinflammatory scarring (rheumatic heart disease) Infective endocarditis Aortic disease Degenerative aortic dilation Syphilitic aortitis Ankylosing spondylitis Rheumatoid arthritis Marfan syndrome

STENOSIS

REGURGITATION



DOESN'T OPEN
PROPERLY



NORMALLY
OPEN



DOESN'T CLOSE
PROPERLY

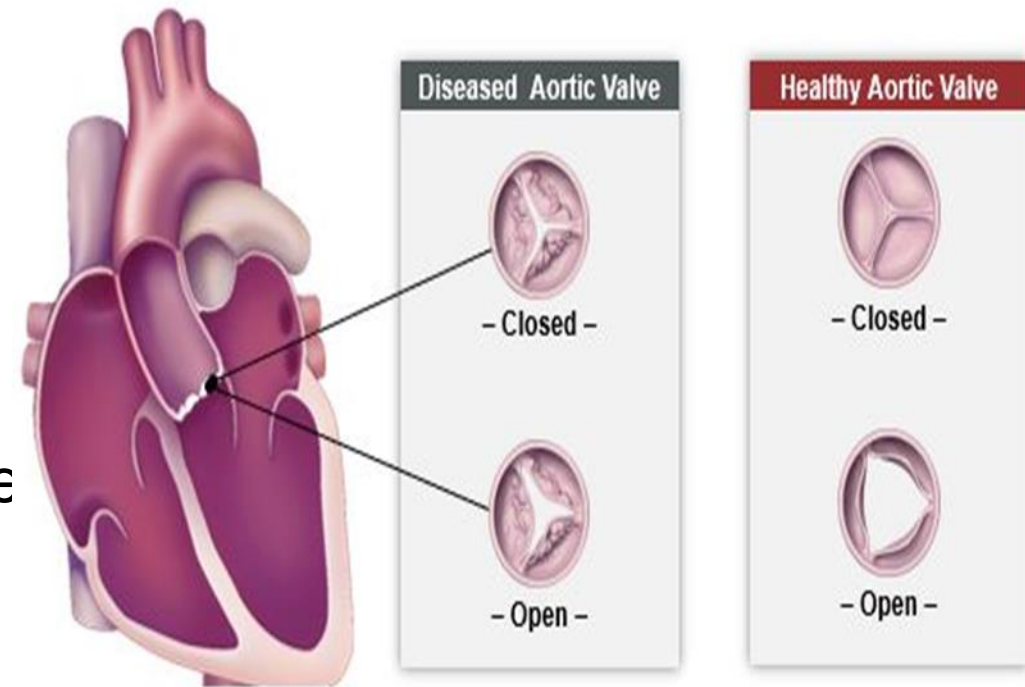


NORMALLY
CLOSED

Aortic Valve Disease

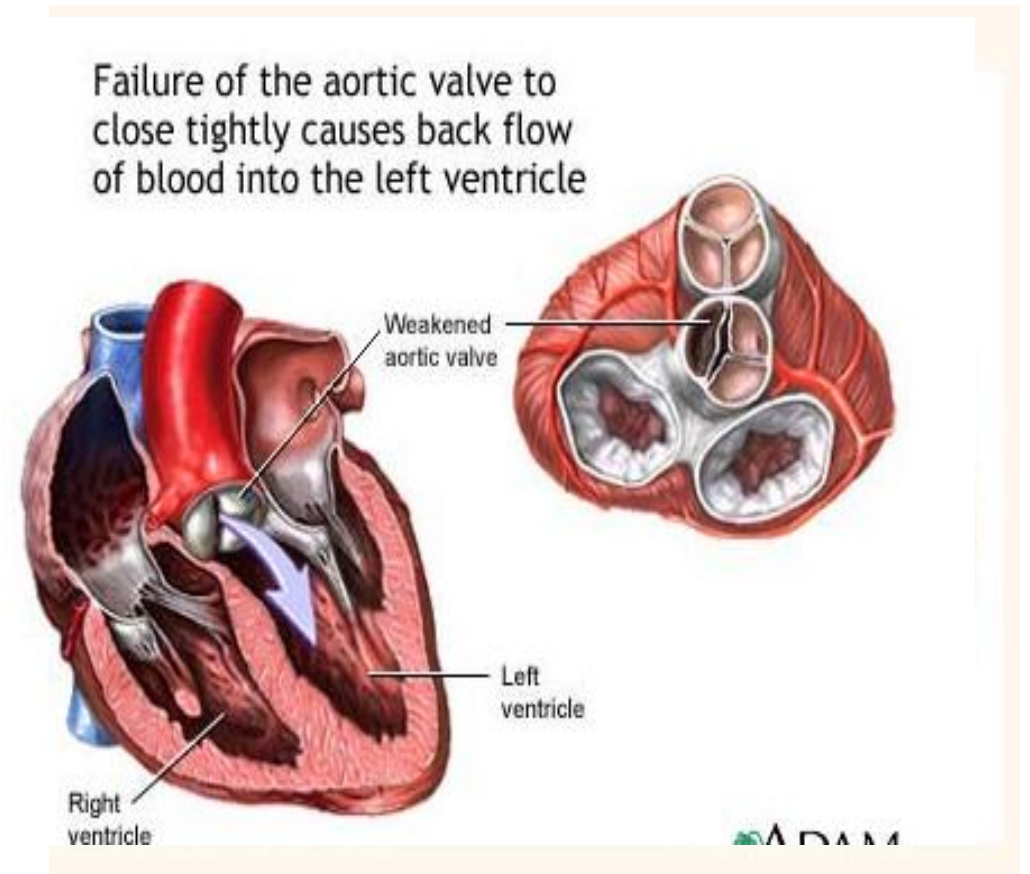
Aortic Stenosis

- Obstruction of blood flow from the left ventricle to the aorta during systole.
- The main causes of aortic stenosis are **idiopathic calcific aortic stenosis** and **rheumatic valve disease**
- There is **concentric** left ventricular hypertrophy
- Inadequate coronary perfusion causes **syncope** and **angina**
- There is a significant risk of sudden cardiac death
- Left ventricular failure eventually develops



Aortic regurgitation

- Retrograde flow of blood from the aorta into the left ventricle during diastole.
- The main causes of aortic incompetence include:
 - post-rheumatic, calcific aortic stenosis
 - dilatation of aortic sleeve
 - infective endocarditis
- It leads to a collapsing pulse and left ventricular failure with dilatation
- There is angina due to inadequate coronary perfusion.
- Left ventricular failure eventually develops

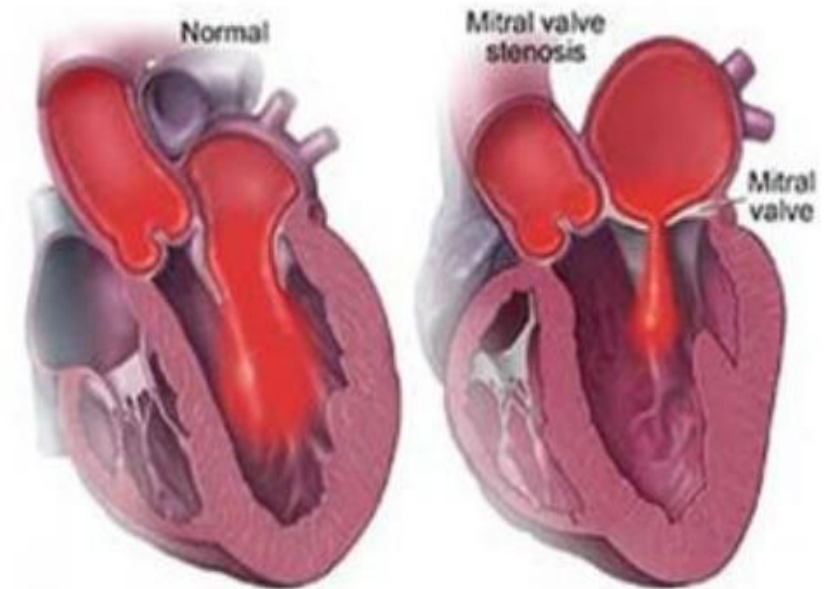


Mitral Valve Disease

Mitral Stenosis

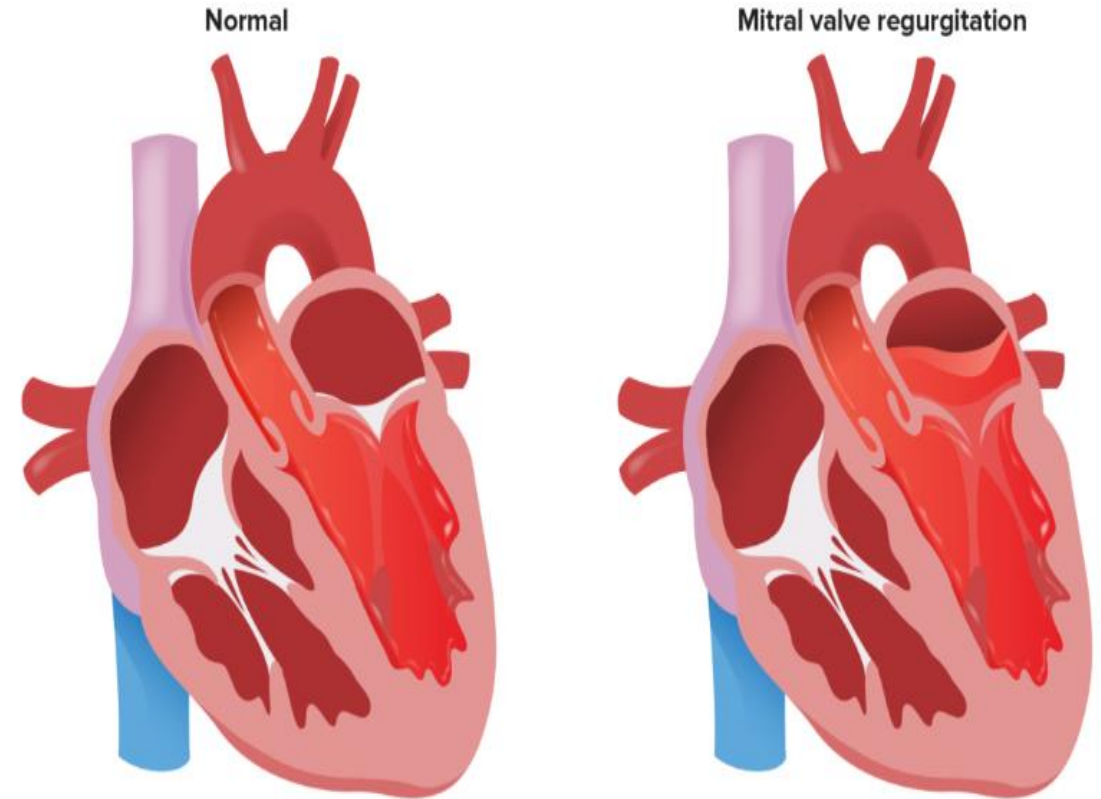
- Obstruction of blood flow from the left atrium to the left ventricle during diastole.
- Mitral stenosis is almost always due to **rheumatic fever**
- There is atrial dilatation, atrial fibrillation, and atrial thrombosis
- It leads to pulmonary venous hypertension and edema
- Pulmonary arterial hypertension causes right ventricular hypertrophy and failure

MITRAL STENOSIS



Mitral Incompetence

- Retrograde flow of blood from the left ventricle into the left atrium during systole.
- The causes of mitral incompetence include:
 - post-rheumatic, infective endocarditis,
 - floppy mitral valve, papillary muscle pathology,
 - valve ring dilatation
- It results in compensatory left ventricular hypertrophy
- There is impaired left ventricular function and failure

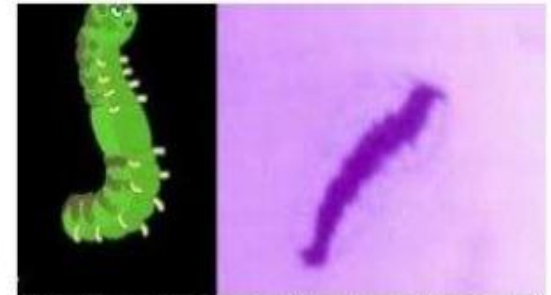


Rheumatic Fever and Rheumatic Heart Disease

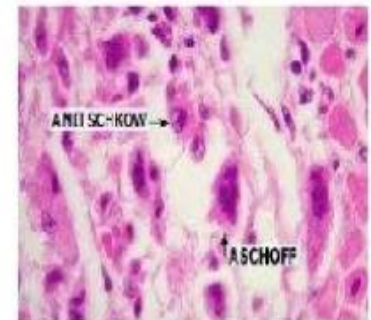
- Rheumatic fever is an acute **multisystem** disease in which there is an immune reaction **2–3 weeks** after a streptococcal pharyngitis
- It affects mainly the joints, subcutaneous tissue, and heart
- In the heart there is a **pancarditis** with involvement of all layers
- The long-term consequences in the **cardiac valves** are pathologically and functionally significant
- There is valve stenosis and/or regurgitation
- The **mitral and aortic** valves are most commonly involved.

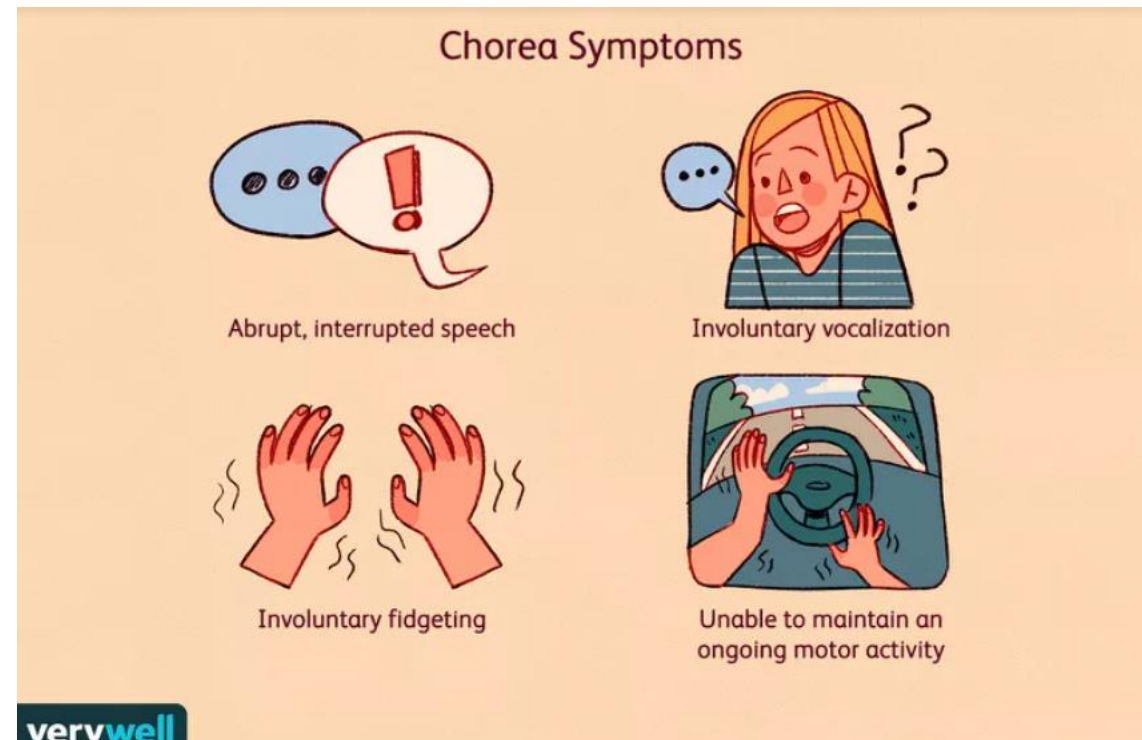
Anitschkow cells (Caterpillar cells) are enlarged macrophages found within granuloma (Aschoff bodies), is associated with rheumatic fever

Larger anitschkow cells may coalesce to form Aschoff giant cells



Anitschkow cell - The caterpillar chromatin pattern of the nucleus in longitudinal section





INFECTIVE ENDOCARDITIS

- Is infection of the valve with formation of thrombotic **vegetations**
- The development of IE requires a sequence of events (**modified Virchow's triad**)
 - **Endothelial Injury, Formation of Non-Bacterial Thrombotic Endocarditis (NBTE), Bacteremia, Colonization**
- Classified into acute and subacute forms
- The major organisms involved include **streptococci(viridans group)** and **staphylococci**
- The **virulence** of the organisms dictates the **severity** of infection and **degree of valve damage**
- Pre-existing valve disease, immunosuppression, and bacteremia are all predisposing factors
- Complications are valve damage, heart failure, septic embolism, toxemia, and glomerulonephritis

INFECTIVE ENDOCARDITIS

Gross Pathology (Macroscopic)

- **Vegetations:** Irregular, friable, bulky masses attached to valve leaflets or endocardium.
 - **Acute IE** (e.g., *S. aureus*): Large, destructive, often ulcerative; may cause rapid leaflet perforation or rupture of chordae tendineae.
 - **Subacute IE** (e.g., *viridans streptococci*): Smaller, firmer, well-organized vegetations; less immediate destruction.
- **Distribution:** Left-sided valves (mitral > aortic) are most common; right-sided (tricuspid) in IV drug users.

INFECTIVE ENDOCARDITIS

Key Clinical-Pathologic Consequences

- **Valvular destruction:** Regurgitation, leaflet perforation, chordal rupture → heart failure.
- **Embolization:** Fragile vegetations embolize → infarcts (brain, spleen, kidney, myocardium), septic emboli, mycotic aneurysms.
- **Immunologic phenomena** (subacute IE):
Glomerulonephritis (immune complex deposition), Osler nodes, Roth spots.

Organism	Clinical Setting	Pathology Characteristic
<i>Streptococcus viridans</i>	Subacute, pre-existing valvular disease	Smaller, less destructive vegetations
<i>Staphylococcus aureus</i>	Acute, aggressive	Large, destructive vegetations; abscesses; rapid valve destruction
<i>Enterococcus faecalis</i>	Elderly, post-instrumentation	Subacute to chronic; often colonize previously damaged valves
<i>HACEK group</i>	Culture-negative subacute	Large vegetations, emboli

Splinter hemorrhages



Janeway lesions

(painless spots on palms/soles of feet)



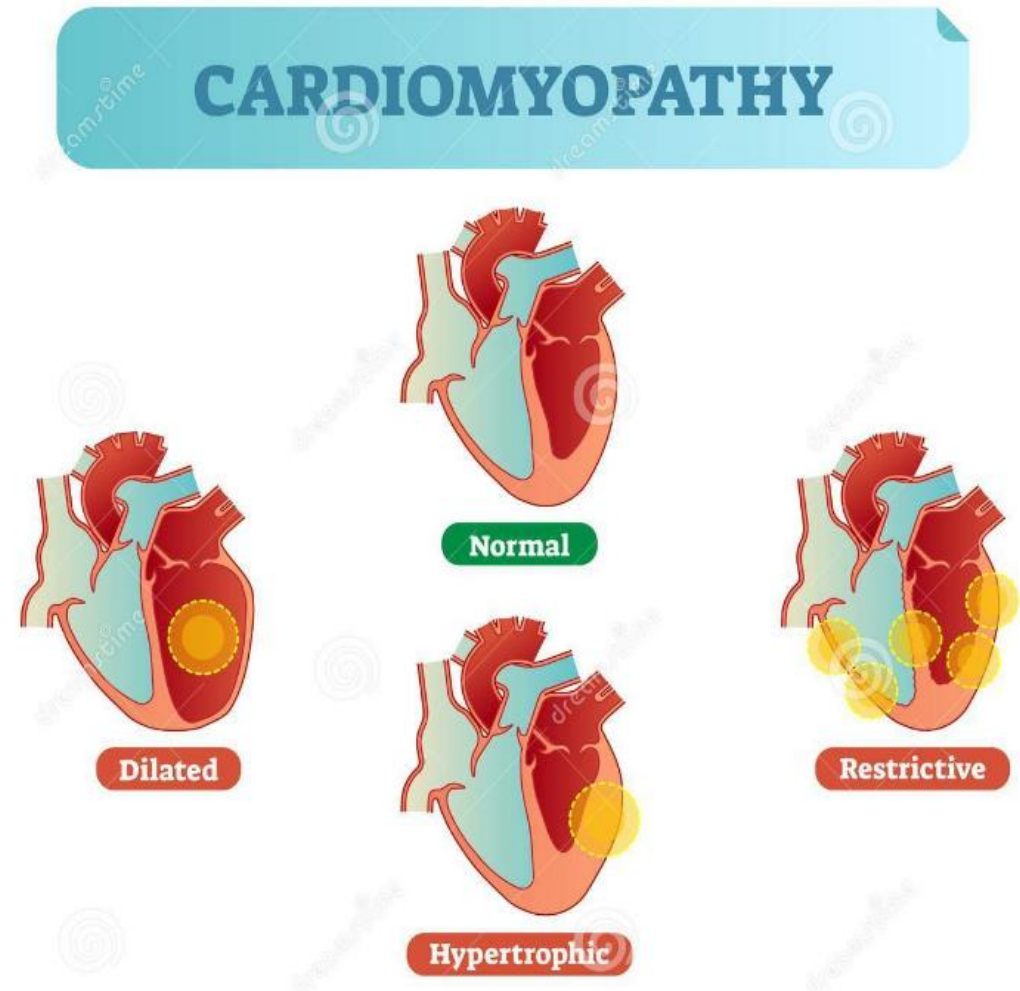
Osler's Nodes

(painful nodules in pulp of fingers/toes)



CARDIOMYOPATHIES

- Myocardial dysfunction that is not the result of coronary artery disease, valve disease, hypertension, or congenital shunts
- The functional abnormality lies within the **myocardium** itself
- The primary cardiomyopathies include
 - **Hypertrophic**: thick-walled left ventricle
 - **Dilated**: dilated poorly contracting ventricle in systole
 - **Restrictive**: the left ventricle fails to relax



Dilated Cardiomyopathy

- Is characterized by ventricular chamber enlargement and **systolic** dysfunction.
- All four chambers are dilated and the walls are **thinned**
- It is the end-stage of many insults to the myocardium.
- There is myocyte stretching and hypertrophy, myocyte damage and diffuse fibrosis.



TABLE 6.6 Causes of dilated cardiomyopathy

Idiopathic	
Post-inflammatory (myocarditic)	Most commonly viral but includes protozoal and bacterial
Genetic	With or without skeletal muscle myopathy Associated with inborn errors of metabolism, e.g. carnitine deficiency or haemochromatosis Sex linked
Nutritional	Severe protein malnutrition
Metabolic	Endocrine disease, e.g. diabetes, hypoparathyroidism, phaeochromocytoma Uraemia
Toxic	Alcohol, drugs, e.g. chemotherapeutic agents Heavy metals, e.g. lead, cobalt, arsenic

Dilated Cardiomyopathy

Clinical Features

- Progressive heart failure
- Dyspnea
- Orthopnea
- Arrhythmias
- Embolic events

Major Causes of Death

- Refractory heart failure
- Ventricular arrhythmias

Pathogenesis

Final common pathway:

- Myocyte injury
↓
Impaired contractility
↓
Chamber dilation
↓
Systolic dysfunction
↓
Heart failure

Hypertrophic Cardiomyopathy

- Hypertrophic cardiomyopathy causes **sudden death**.
- There is left ventricular hypertrophy (usually asymmetrical), especially involving the **interventricular septum**.
- There is **systolic** and **diastolic** dysfunction.
- Many cases are due to myocyte contractile protein gene mutations, especially of the myosin heavychain.
- There is myocyte hypertrophy and **disarray** with variable diffuse fibrosis.



Normal heart
(cut section)



Hypertrophic
cardiomyopathy

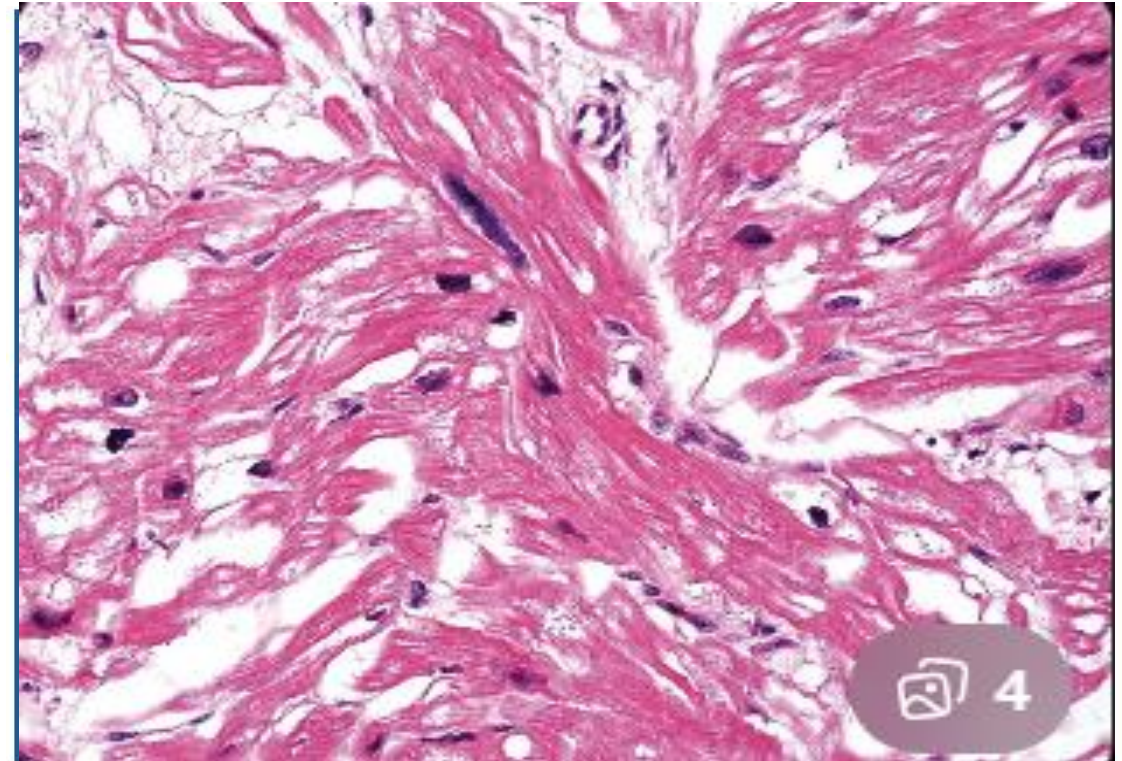
Hypertrophic Cardiomyopathy

Clinical Features

- Dyspnea
- Syncope
- Angina
- Sudden death

Sudden Cardiac Death

- Especially in:
 - Young athletes
 - Adolescents
 - Usually due to ventricular arrhythmias.



Arrhythmogenic Right Ventricular Cardiomyopathy

- Is a rare inherited heart-muscle disease that is a cause of **sudden death in young people and athletes**.
- It is characterized by progressive replacement of the right ventricle myocardium by either segmental or diffuse **fibrofatty tissue**
- The fibrofatty replacement leads to **ventricular arrhythmias**.
- Segments of the right ventricle show dilatation and thinning of the wall with fatty replacement of the muscle

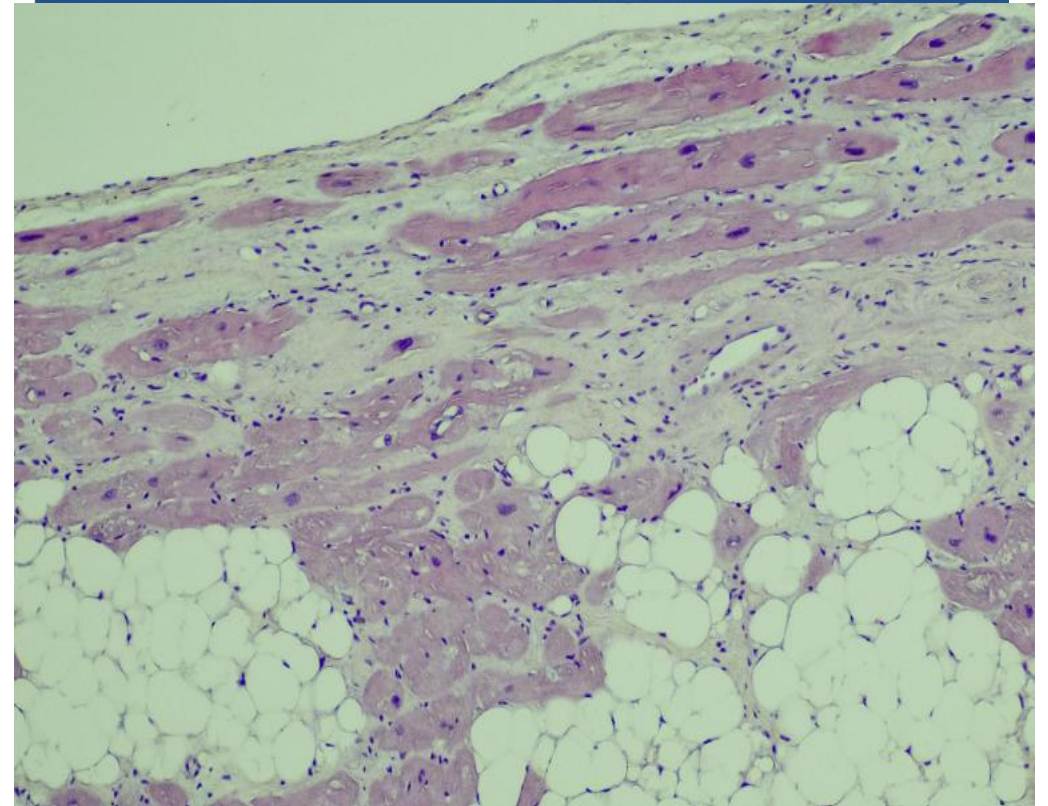
Arrhythmogenic Right Ventricular Cardiomyopathy

Microscopy

- Characteristic:
 - Fibrofatty replacement
 - Myocyte loss
 - Chronic inflammation

Clinical Features

- Palpitations
- Syncope
- Ventricular tachycardia
- Sudden death

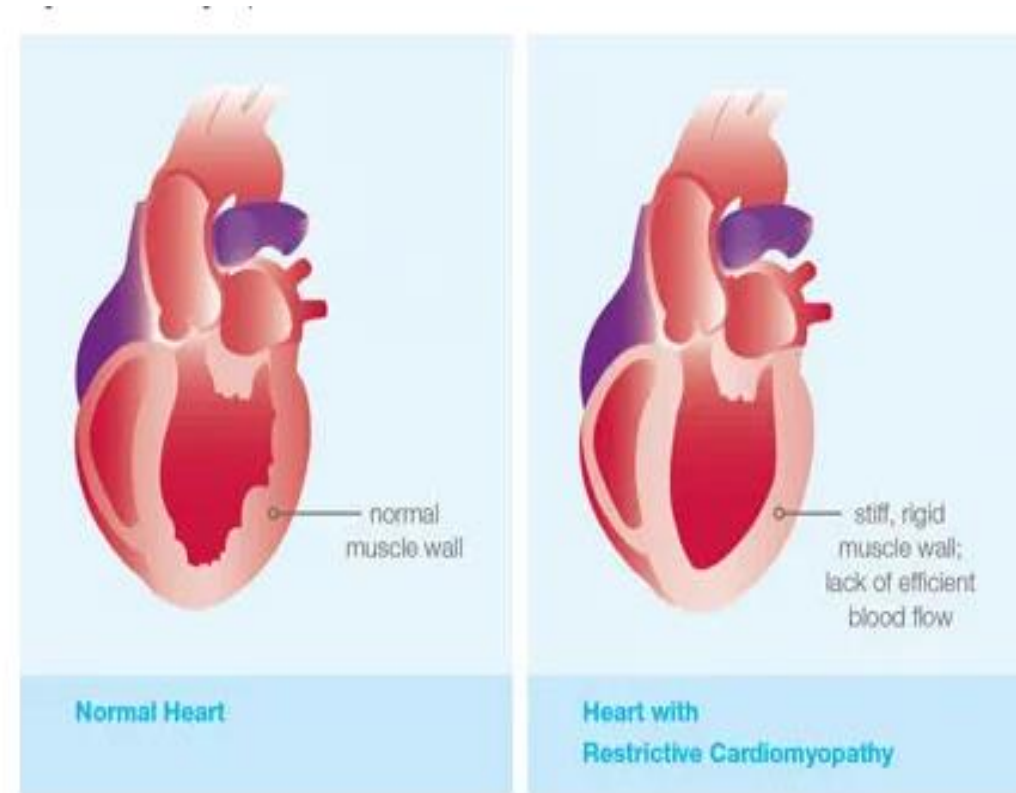


Restrictive Cardiomyopathy

- Cardiomyopathy characterized by impaired ventricular filling due to decreased ventricular compliance.
- Systolic function initially preserved.

Causes

- Amyloidosis (Most Common)
- Endomyocardial Fibrosis
- Radiation Injury
- Sarcoidosis
- Hemochromatosis
- Storage Disorders



Restrictive Cardiomyopathy

Pathogenesis

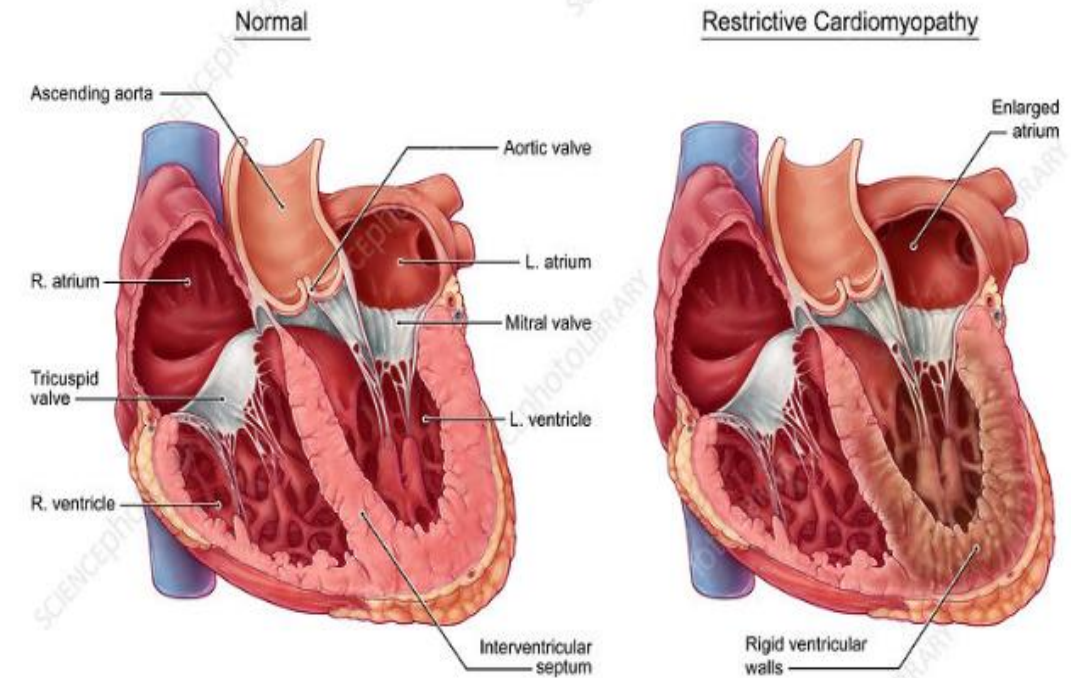
- Rigid ventricular wall
 - ↓
- Impaired diastolic filling
 - ↓
- Elevated atrial pressure
 - ↓
- Heart failure

Gross Pathology

- Ventricles normal size
- Biatrial dilation common
- Thickened myocardium (depending on cause)

Clinical Features

- Right-sided heart failure
- Peripheral edema, Ascites
- Exercise intolerance



Takotsubo (Stress) Cardiomyopathy

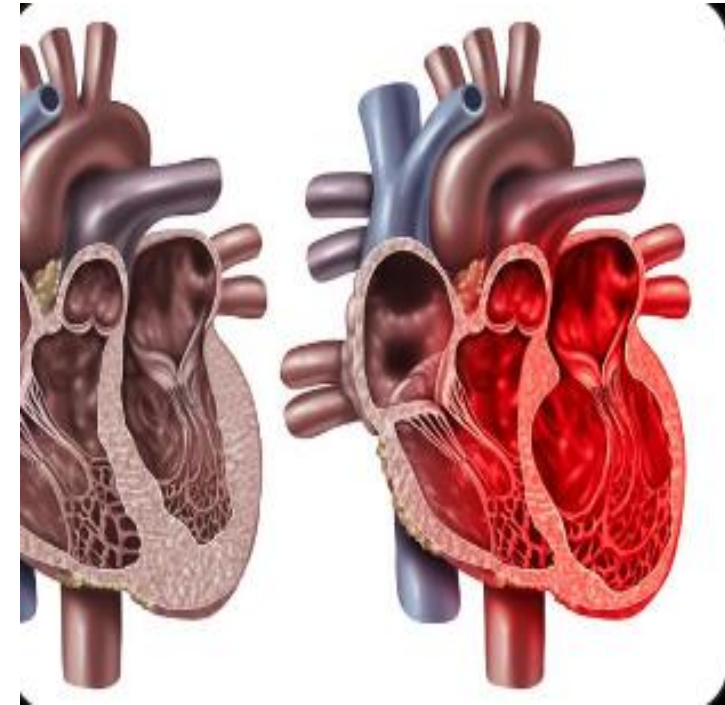
- Transient reversible cardiomyopathy triggered by emotional or physical stress.
- Also called: **Broken Heart Syndrome**

Pathogenesis

- Catecholamine surge causes:
 - Myocardial stunning
 - Transient LV dysfunction

Gross Findings

- Apical ballooning of LV.
- Shape resembles Japanese octopus trap: "Takotsubo"



Myocarditis

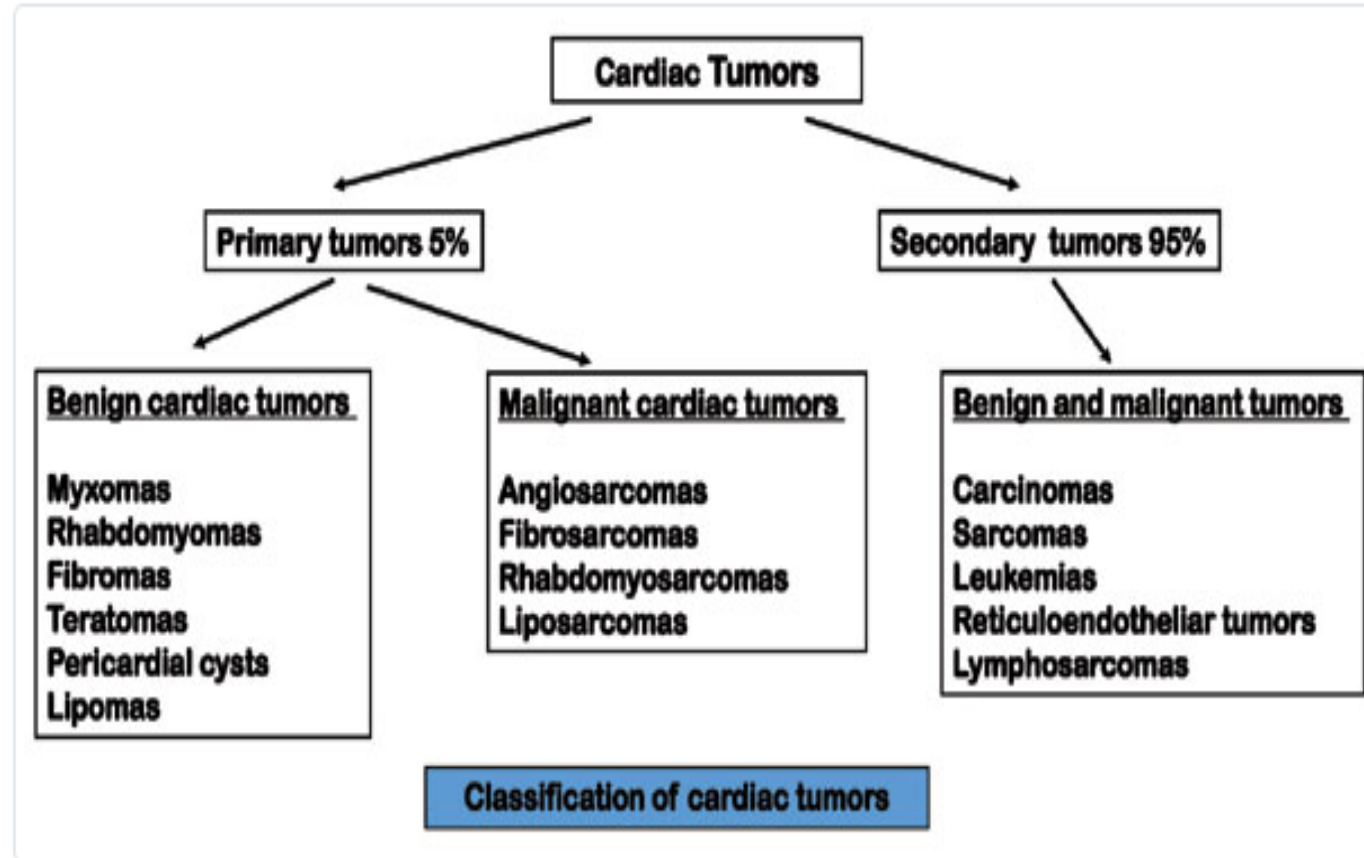
- A major cause of acquired myocardial dysfunction.

Causes

- **Viral:** Most common.
 - Coxsackie virus B
 - Adenovirus
 - Parvovirus B19
- **Other Causes**
 - Chagas disease
 - Autoimmune disorders
 - Drug hypersensitivity

TUMORS OF THE HEART AND PERICARDIUM

- Primary cardiac and pericardial tumors are rare
- Most primary cardiac tumors are benign, with only 25% being malignant
- The most common primary cardiac tumor is the **cardiac myxoma**
 - usually occurs in adult life in the left atrium.
 - It forms a gelatinous mass up to 6 cm in diameter
 - It may create intermittent ball valve obstruction of the mitral valve
 - It may also present with embolic phenomena due to fragmentation
- In children, **rhabdomyomas** are the most common cardiac tumor
 - These tumors tend to involve the left ventricle
- In adults, metastatic malignant tumors are the most frequent tumors to involve the heart and pericardium



THANK YOU!!!